An Interpretable Multi-Task Fusion Framework for Wind Farm Siting

Combining Policy Semantics and Meteorological Features

— A Case Study over 73–135°E, 37–50°N, China

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Abstract

Wind farm siting requires balancing meteorological resources, engineering feasibility, and regulatory compliance. Existing approaches often reduce policy constraints to binary masks, overlooking nuanced signals—such as encouragement or prohibition—embedded in regulatory texts.

We propose a dual-stream cross-modal learning framework that jointly encodes policy semantics and geospatial numerical features. A policy semantic stream extracts dense province-level representations from Chinese regulatory documents using parameter-efficient fine-tuning of a pretrained language model, while a numerical feature stream encodes heterogeneous meteorological, topographic, and infrastructure data. Cross-modal attention allows numerical features to query policy semantics, and a gating module adaptively fuses the streams. Multi-task learning jointly optimizes site suitability, wind resource potential, and engineering compliance.

Experiments over 13,197 ERA5 grid points in northern China show that integrating policy semantics improves siting performance under constrained negative sampling (Experiment 4, 489 test samples, 33.3% positives): the proposed model reaches AUC-ROC of 0.8386, exceeding the no-policy baseline by ΔAUC = +0.0441 (full vs. zeroed policy stream). After controlling for the mean-policy (province-fixed-effect) baseline, the additional contribution from learnable policy semantics is ΔAUC = +0.0004 (ΔAP = +0.0028; paired-bootstrap two-sided $p = 0.94$ on n=2000 sample-level resamples), indicating that the policy stream's learned encoder contributes essentially no incremental discriminative power over and above what the per-province mean policy vector already encodes—the policy stream therefore operates as a *province-level prior corrector* rather than a fine-grained semantic encoder. Independent validation against six SCADA wind farms (the only ones for which Chen [22] SCADA data are available; the validation file additionally lists three synthetic duplicates that lack measured power data) shows that 2/6 sites rank in the top 10% and 3/6 in the top 15% of all 13,197 grid points (mean rank 3160). Per-province macro-averaged AUC over 11 provinces with both pos and neg samples is 0.7494 (95% bootstrap CI over province resamples: $[0.58, 0.87]$), exposing substantial inter-provincial variability that the overall AUC of 0.8444 on the 9.6%-positive realistic test set partially hides. A RandomForest baseline on the same features scores AUC = 0.8838 (AP = 0.7952), exceeding the proposed model's AUC by 0.045 ($p < 0.001$ over 2000 bootstrap resamples), highlighting that architectural complexity does not by itself translate into better ranking on this task. High-suitability regions align with national development plans, concentrated in Xinjiang, Inner Mongolia, Gansu, and Jilin.

Our results demonstrate that learnable policy representations enable more realistic, policy-aligned wind farm siting, offering a new paradigm for data-driven energy planning under regulatory constraints.

Keywords: wind farm siting; language models; cross-modal attention; multi-task learning; China

Contents

1 Introduction

Wind farm siting has traditionally been formulated as a spatial optimisation problem balancing resource potential, engineering feasibility, and environmental constraints; cf. [47] for a textbook treatment of wind-resource theory, turbine engineering, and grid integration. However, regulatory policies increasingly shape where and how renewable-energy projects can be developed, yet policy information remains largely absent from data-driven siting models.

By end-2024, China's total installed wind capacity reached 521 GW, accounting for 45.8% of the global total [18, 24]; combined wind and solar capacity surpassed coal-fired power for the first time in early 2025 [24]. The 14th and 15th Five-Year Plans for renewable energy [7, 8] both mandate substantial further expansion. Current development faces compounding constraints: wind resources are concentrated in the northwest, north, and northeast, while consumption centres lie on the eastern seaboard; ecological red lines, permanent basic farmland, and military exclusion zones add rigid spatial constraints; and differentiated provincial and municipal policies further raise the compliance burden [19]. Recent deep-learning assessments such as [54] quantify China's undeveloped barren-land wind potential at ~11,108 TWh/year under land-use constraints, underscoring the magnitude of the siting problem that data-driven methods must address. Existing siting frameworks handle numerical spatial features and unstructured policy text as separate concerns, leaving a structural gap.

Existing approaches fall into three categories. Traditional GIS-based multi-criteria decision analysis relies on expert-assigned criterion weights, making results sensitive to the weighting scheme and unable to ingest policy-document semantics directly [4, 20]. Pure machine-learning frameworks have advanced discriminative accuracy — Bilgili et al. [26] achieved 96.07% accuracy with XGBoost and SHAP (SHapley Additive exPlanations), and Chen et al. [1] built spatio-temporal probability models at China's regional scale; Sun et al. [51] likewise applied four XAI algorithms (DNN, RF, XGBoost, LightGBM) with SHAP-based interpretation across China, USA, and EU, and Gao et al. [55] used deep reinforcement learning to optimise onshore wind-station locations in Guangdong — but all of these works treat regulatory compliance as a binary preprocessing mask, discarding the semantic gradient among encouraging, restrictive, and prohibitory intents. The recent proliferation of systematic reviews [23, 44] documents this standardisation gap and motivates models that preserve policy context rather than discarding it. NLP methods open a path for extracting constraints from unstructured text: Buster et al. [2] used BERT to extract setback ordinances from 2,500+ US county zoning documents, and Li et al. [3] quantified policy-support intensity in Chinese energy texts. None of these works, however, fuses encoded results with spatial numerical features into a unified trainable siting model.

The shared limitation of all three approaches points to the same gap: how to let policy-text semantics genuinely participate in spatial decisions within a single trainable architecture, rather than remaining at the rule-filtering preprocessing stage. To address this gap, we propose a dual-stream cross-modal learning framework and, in doing so, arrive at three interconnected contributions.

First, we move beyond the conventional treatment of regulatory constraints as binary exclusion masks by introducing policy semantics as learnable representations within a jointly trainable architecture, enabling regulatory information to participate directly in data-driven siting decisions rather than operating as a preprocessing filter.

Second, we develop a dual-stream cross-modal learning framework that integrates policy-text semantics with heterogeneous geospatial features, enabling dynamic interactions between regulatory information and local resource conditions within a unified trainable model.

Third, our analysis exposes a structural property of the proposed architecture: because the policy stream produces a single embedding vector per province, its contribution to suitability operates as a provincial-level prior correction rather than a grid-level discriminative feature. The ablation evidence (ΔAUC = +0.0441 for full vs. zeroed policy stream on the constrained Experiment 4 test set) and the near-zero point-level SHAP values (< 0.03) are jointly explained by this resolution constraint. Critically, once province fixed effects are controlled—by zeroing the fine-grained semantic content while keeping the mean policy vector per province—the residual semantic contribution shrinks to ΔAUC = +0.0004 (paired-bootstrap $p = 0.94$, statistically indistinguishable from Full), demonstrating that the entire policy-stream benefit is province-prior correction rather than learned fine-grained semantics. This finding clarifies the boundary conditions under which semantic policy features can complement numerical siting models in hierarchically governed spatial systems.

The remainder of the paper is organised as follows. Section 2 reviews three families of related work—GIS-based multi-criteria decision analysis, machine-learning siting, and NLP for regulatory text—and identifies the systematic gaps that motivate a cross-modal design. Section 3 describes the study area, the ERA5/OSM/DEM feature stack, the policy corpus, and the OSM-turbine positive-sample construction, together with the spatial-block negative sampling protocol that avoids autocorrelation leakage. Section 4 details the dual-stream cross-modal architecture, the TabTransformer numerical stream, the BERT+LoRA policy stream, the cross-modal attention fusion module, the three-task output heads, and the four-experiment protocol under which all reported numbers can be reproduced from a single saved checkpoint. Section 5 reports the ablation, cross-provincial macro-AUC, tree-based baseline, SHAP, and independent SCADA-site validation results. Section 6 discusses the unified explanation of policy-stream behaviour and SHAP gate values, the geographic generalisation boundary, and the methodological implications for future hierarchical policy-aware siting models. Section 7 concludes. An appendix documents the four-stage iterative protocol correction that produced the final experimental configuration.

2 Literature Review

Wind farm siting research has evolved along three technical lines: MCDM for policy-weight modelling, machine learning for numerical-feature mining, and NLP for regulatory-text semantic extraction. Each has mature applications, but none has provided a satisfactory answer to how all three information types can be unified in a single trainable framework.

2.1 Multi-Criteria Decision Analysis

MCDM is the most widely used traditional framework for wind farm siting, primarily encompassing the Analytic Hierarchy Process, the Technique for Order of Preference by Similarity to Ideal Solution, and their fuzzy extensions [4, 20]. These methods assemble weighted scoring systems from criteria such as wind speed, terrain slope, grid distance, and land-use type. Their strength lies in transparency and ease of GIS integration, which has made them popular for provincial and regional siting studies [48].

Two fundamental limitations constrain this framework. First, weight subjectivity: different experts assign significantly different importance to the same criterion, and the effect of the weighting scheme on final suitability scores is non-negligible [4, 20]. Second, semantic loss in policy representation: policy constraints are reduced to binary masks or discrete categorical scalars, relying entirely on the researcher's subjective reading of policy text and unable to capture the semantic gradient among encouraging, restrictive, and prohibitory regulatory intents.

2.2 Machine Learning Methods

Machine learning methods have substantially advanced discriminative accuracy beyond the linear-weighting limitation of MCDM. Bilgili et al. [26] achieved 96.07% accuracy with XGBoost combined with SHAP on a Turkish wind-farm dataset. Chen et al. [1] built spatio-temporal probability distribution models for wind and solar siting at China's regional scale. Recent reviews [23] consolidate progress on machine-learning siting and short-term forecasting across wind, solar, and hybrid renewable systems.

A shared structural limitation across these works is that regulatory compliance is treated as a binary preprocessing mask applied before model training, rather than as a learnable feature. This approach discards the contextual semantic information in Chinese energy-policy language. Furthermore, most machine-learning siting studies lack independent validation against real operational data, relying instead on administrative boundary labels or expert judgements, which introduces circularity. Methodological surveys of explainable AI [41] and recent benchmarking work on tabular deep learning [39, 40] together provide the contemporary methodological vocabulary on which our cross-modal design rests.

2.3 NLP and Large Language Model Methods

NLP provides a new pathway for extracting siting constraints from unstructured policy text. Buster et al. [2] used BERT to automatically extract wind-turbine setback regulations from over 2,500 US county zoning documents, achieving 85–90% accuracy. Li et al. [3] applied BERT classifiers to Chinese energy policy texts to automatically quantify policy-support intensity.

Two key limitations constrain this direction. First, spatial mapping is absent: results are output as reports or statistical summaries, not mapped to candidate site grid points. Second, joint training is absent: policy-text encoding and spatial numerical feature fusion is completed through manual post-processing rather than jointly optimised in a unified trainable architecture. GPT-4-class methods also face high inference costs and non-reproducibility; recent benchmarks show LLMs still struggle with complex regulatory reasoning in environmental permitting contexts [25]. By contrast, BERT+LoRA parameter-efficient fine-tuning retains pre-trained language knowledge at low inference cost and has proven competitive on text-numerical fusion tasks in tabular domains [32], motivating its adoption here. semantic knowledge while adapting to domain-specific tasks at minimal computational cost [12], making it better suited for building a reproducible unified siting framework.

2.4 Research Gaps

Table 1 summarises six dimensions of systematic gaps in existing wind-farm siting research. MCDM methods are mature in spatial analysis but rely on hand-crafted scalar policy representations; machine-learning methods lead in numerical feature mining but reduce policy information to a post-processing mask; NLP methods have achieved breakthroughs in semantic extraction but lack integration with spatial numerical features. The complementary strengths of all three approaches point towards the core design motivation of this study: a unified framework that simultaneously encodes numerical features and policy-text semantics and fuses them through learnable cross-modal attention.

Table 1: Comparison of existing methods and the proposed framework

Dimension	Existing research	This work
Policy modelling	Qualitative or binary mask	BERT+LoRA dense semantic vectors, provincial spatial mapping
Numerical features	GIS single-modal, expert weights	TabTransformer (16 ERA5+DEM+OSM features)
Fusion mechanism	None (independent streams or rule concatenation)	Cross-attention + gated fusion
Multi-task learning	Single-objective optimisation	GradNorm three-task heads
Ground-truth labels	Expert judgement or administrative boundaries	OSM turbine GPS clustering + independent validation
Spatial coverage	Provincial or regional scale	National scale (13,197 grid points, 0.1°)

The gap analysis motivates the design choices taken in the remainder of the paper. Rather than replacing MCDM, machine learning, or NLP in their respective strengths, the present framework retains the discriminative power of tree- and attention-based learners, the unstructured-text capability of pre-trained language models, and the policy-semantic expressiveness of dense regulatory embeddings, and unifies them through learnable cross-modal attention. The next section introduces the data stack, the spatial sampling protocol, and the OSM-derived positive labels that make nationwide supervised training feasible.

3 Study Area and Data

The data system consists of four components: meteorological features, terrain features, engineering features, and a policy corpus, covering 13,197 ERA5 land grid points over 73–135°E, 37–50°N.

3.1 Study Area

At 0.1° resolution, 13,197 ERA5 land grid points cover key wind-energy provinces including Xinjiang, the Gansu Hexi Corridor, the Inner Mongolian grasslands, Heilongjiang, and the Shandong Peninsula. Figure 1 shows the spatial distribution of ERA5 annual-mean 100 m wind speed; nine independent validation wind-farm sites are marked; the Kashgar site is excluded due to an ERA5 correlation coefficient of only 0.154, insufficient for reliable coordinate inference.

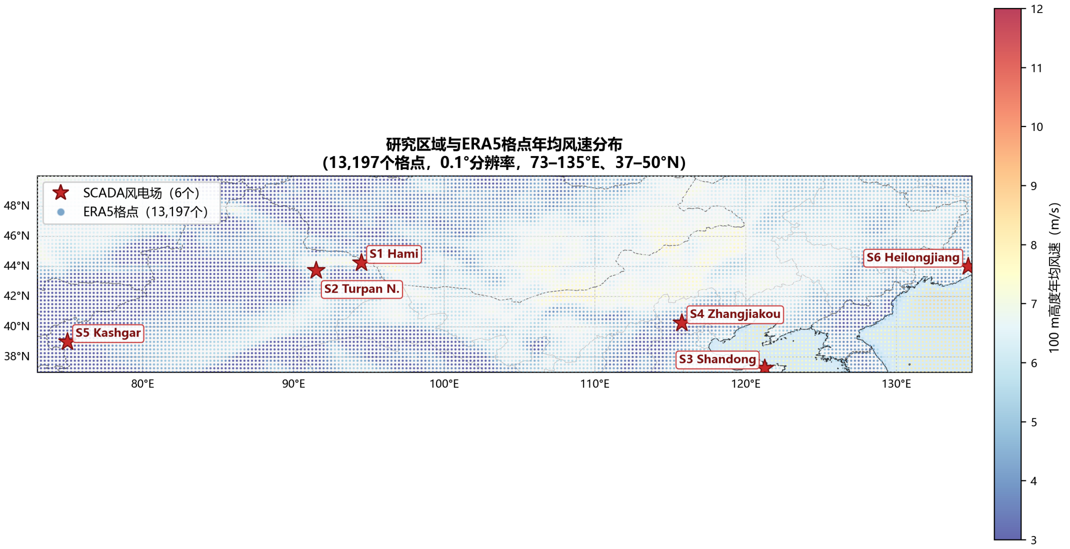

Figure 1: Study area and ERA5 annual-mean 100 m wind speed distribution

3.2 ERA5 Meteorological Features

Nine wind-energy features are extracted from ERA5 reanalysis data at 100 m height for 2018–2025 (ERA5 downscaling uncertainties over complex terrain are discussed in Ding and Hsieh [27]): annual mean wind speed, wind speed standard deviation, wind power density, 75th percentile wind speed, annual effective generation hours (wind speed >3 m/s), annual high-efficiency generation hours (wind speed >10 m/s), seasonal coefficient of variation, Weibull k parameter, and Weibull c parameter.

3.3 Terrain Features

Four terrain features are extracted from a 0.1° resolution DEM, with no missing values across all 13,197 grid points (Figure 2). Mean elevation is 1,154 m (max 5,327 m); mean slope is 0.86° (max 20.8°); the topographic position index ranges from −2,057 to +1,955 m; and mean terrain roughness is 419 m (max 6,026 m).

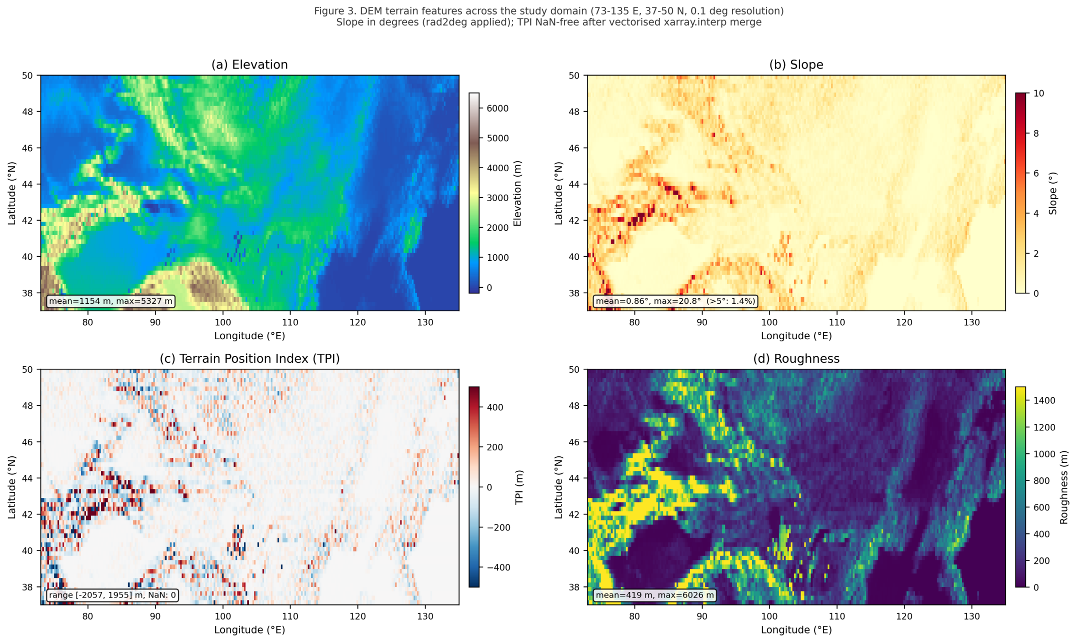

Figure 2: DEM terrain features over the study area

3.4 Engineering and OSM Features

Three engineering accessibility features are extracted from OpenStreetMap [10]: distance to the nearest transmission line, road, and settlement, jointly reflecting grid connection costs, construction accessibility, and social impact constraints. Engineering compliance at candidate sites is constrained by the national standard GB/T 19963.1–2021 [31], which prescribes the minimum technical requirements for connecting onshore wind farms to the power system and is enforced during grid permitting. A composite LCOE proxy index is constructed as the 16th feature to approximate the levelised cost of energy. Shorter distance to transmission lines, flatter terrain, and shorter distance to roads all reduce the LCOE proxy, indicating better engineering feasibility.

3.5 Policy Corpus

We collected 81 wind-energy and renewable-energy policy documents: 67 national-level documents and 14 provincial planning documents from six key provinces (Xinjiang, Inner Mongolia, Gansu, Jilin, Ningxia, and Shandong), spanning 2002–2026. Manual annotation of national documents yielded 1,400 sentences (471 neutral, 553 encouraging, 251 restrictive, 125 prohibitory; inter-annotator $\kappa = 0.81$); rule-based annotation of provincial documents added 12,759 sentences, giving a combined corpus of 16,433 sentences. Provincial documents are dominated by encouraging clauses, with the encouraging ratio varying across provinces: Ningxia 5.0%, Inner Mongolia 4.7%, Gansu 4.1%, Shandong 2.3%, Jilin 1.4%, Xinjiang 0.6%.

Although the corpus spans 2002–2026 and includes the 15th Five-Year Plan signals, this does not introduce temporal leakage into the 2019–2020 validation for two reasons. First, the policy stream operates as a provincial-level prior corrector (provincial prior correction, Section 4.4) rather than a point-level discriminator: its contribution is structurally insensitive to the addition of forward-looking documents that reinforce rather than reverse existing provincial development priorities. Because all grid points within a province share an identical policy embedding vector, any marginal change in that vector from adding a continuity document produces only a uniform shift in provincial suitability priors, not a discriminative signal between individual grid points. Second, the 15th Plan extends and formalises long-term development trajectories—particularly large-base construction in the Three-North region—that were already established policy direction before the validation sites were built. The inclusion of forward-looking documents therefore reflects the *continuity* of China's wind energy governance rather than ex-post information.

Figure 3: Annual trends of the Chinese wind-energy policy corpus (2002–2026)

3.6 Training Samples and Negative Sample Generation

Positive samples were derived from 103,568 OSM wind turbine GPS coordinates clustered with DBSCAN (Density-Based Spatial Clustering of Applications with Noise) [15] ($\varepsilon = 3\,\text{km}$, $min\_samples = 3$), yielding 1,115 wind farm centroids. These were split into training (789), validation (163), and test (163) sets using the Provincial Stratified Split

(PSS) protocol, which ensures each of the 10 representative provinces is represented in all three subsets.

Negative samples were drawn from a constrained pool of grid points satisfying four criteria: (1) spatial buffer exceeding 10 km from all positive samples, to prevent spatial autocorrelation leakage whereby adjacent grid points could be trivially distinguished from positives by proximity alone [42]; (2) terrain slope $\leq 25°$, to exclude physically infeasible development terrain where wind farm construction is structurally impractical; (3) elevation $\leq$ 3,500 m, to exclude high-altitude areas inconsistent with the deployment envelope of current commercial turbine technology; and (4) distance to nearest settlement $\geq$ 0.5 km, to avoid grid points whose settlement proximity would create trivially identifiable non-sites. The constrained pool contained 12,294 eligible grid points. Negatives were sampled using province-stratified sampling matching the positive sample provincial distribution, yielding 789 training, 163 validation, and 163 test negative samples respectively, maintaining a 1:2 positive-to-negative ratio throughout all data partitions.

This constrained sampling protocol ensures that negative samples represent genuinely ambiguous siting candidates—grid points with adequate terrain and distance characteristics that were nonetheless not developed—rather than trivially unsuitable locations. The resulting evaluation is therefore a meaningful test of the model's ability to discriminate between viable and developed sites, not merely between extreme terrain and operational wind farms.

Figure 4 shows 789 positive training samples (OSM wind-farm clusters) and 1,578 negative samples. The nine validation sites are distinguished by confidence level: Sites 1, 2, 6, and 7 have Pearson $r \geq 0.50$ (high confidence); Sites 3, 5, 8, and 9 have r between 0.30 and 0.50 (moderate confidence); Site 4 has $r = 0.240$ (lower confidence); the Kashgar site ($r = 0.154$) is excluded. The mean correlation coefficient across the nine sites is $\bar{r} = 0.454$.

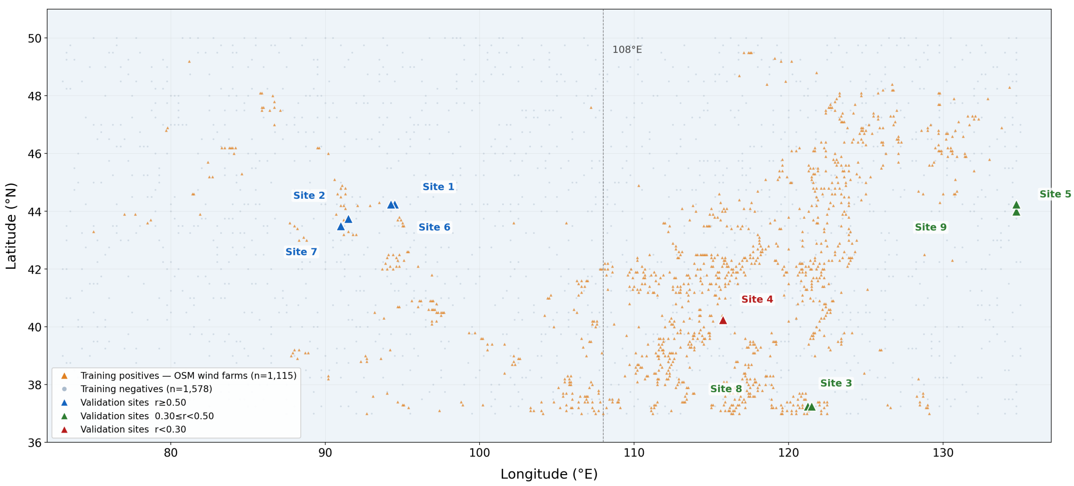

Figure 4: Spatial distribution of training samples

The final data partitioning scheme and feature construction pipeline adopted here were established through a four-stage iterative validation protocol designed to systematically eliminate spatial autocorrelation leakage, feature scaling artefacts, and settlement proxy bias prior to any model training; the full protocol is documented in Appendix A.

4 Methodology

The proposed dual-stream cross-modal fusion architecture consists of two parallel encoding streams, one cross-modal attention fusion module, and three multi-task output heads. We frame cross-modal fusion in the broader taxonomy of Baltrušaitis et al. [33], where numerical and policy modalities interact through a late attention-based fusion stage that conditions one modality's representation on the other. The numerical-feature stream captures meteorological, terrain, and engineering constraints; the policy-semantic stream encodes regulatory text into dense vectors that interact with numerical features; the fusion module combines both streams through attention and gating; and the output heads jointly optimise three siting objectives.

4.1 Overall Framework

Figure 5 shows the complete dual-stream architecture. The left blue stream encodes 16 structured spatial features with TabTransformer, outputting $\mathbf{F}_{\text{num}} \in \mathbb{R}^{64}$. The right green stream encodes 81 policy documents with BERT+LoRA and maps embeddings to 13,197 grid points, outputting $\mathbf{F}_{\text{policy}} \in \mathbb{R}^{64}$. The central purple module performs cross-modal attention using $\mathbf{F}_{\text{num}}$ as query and $\mathbf{F}_{\text{policy}}$ as key and value, then adaptively fuses both streams via a Sigmoid gate to produce $\mathbf{F}_{\text{fused}} \in \mathbb{R}^{64}$. Three GradNorm-balanced task heads output the site suitability map, wind-resource potential map, and engineering suitability map.

The **numerical-feature stream** encodes 16 structured spatial features via TabTransformer [5]. The core motivation for TabTransformer is that the 16 features are highly heterogeneous — meteorological, terrain, and engineering dimensions differ enormously in scale and exhibit nonlinear interactions (e.g., the joint cost constraint of slope and elevation). TabTransformer applies learnable linear embeddings per feature followed by Transformer self-attention, automatically capturing cross-feature contextual dependencies without manual feature engineering; recent deep-learning surveys of tabular data [39, 40] and the foundational treatment of representation learning [46] together motivate this attention-based design choice over hand-engineered concatenation. The 16 features are mapped to a 64-dimensional latent space, processed by 2-layer 4-head self-attention [13], and mean-pooled to produce $\mathbf{F}_{\text{num}}$.

The **policy-semantic stream** encodes unstructured regulatory text into dense vectors that interact with numerical features. We fine-tune on 1,400 annotated sentences with BERT+LoRA (Low-Rank Adaptation) [11, 12, 34, 36], using `hfl/chinese-roberta-wwm-ext` as the backbone and injecting low-rank adaptation matrices into the Q and V projection matrices. The LoRA rank is set to 16 with a scaling factor of 32; these values were selected via a grid search over ranks $\{4, 8, 16, 32\}$ and scaling factors $\{8, 16, 32, 64\}$ on the validation set macro-F1, with rank=16 and scaling=32 yielding the highest macro-F1 of 0.850 while keeping trainable parameters at approximately 0.3% of the full model. Compared with full fine-tuning, LoRA at this configuration compresses trainable parameters effectively, preventing overfitting given the limited corpus size [30]; recent benchmarks confirm that in data-scarce settings PEFT is a prerequisite for robust generalisation rather than merely an efficiency choice. The fine-tuned model generates sentence-level embeddings per document, aggregates by provincial boundary, and maps each of the 13,197 ERA5 grid points to $\mathbf{F}_{\text{policy}} \in \mathbb{R}^{64}$.

The **cross-modal attention fusion module** adaptively combines both stream outputs using 4-head cross-modal attention ($d_{\text{model}} = 64$), with $\mathbf{F}_{\text{num}}$ as query and $\mathbf{F}_{\text{policy}}$ as key and value, followed by residual connection to yield $\mathbf{F}_{\text{cross}}$. Our use of directional pairwise cross-modal attention follows the canonical Multimodal Transformer formulation [33, 43] and is conceptually adjacent to recent image-domain Transformer encoders [50]; the domain- adaptation perspective on shared-embedding alignment is discussed in [45, 49]. A Sigmoid gate then fuses cross-modal and original numerical features into $\mathbf{F}_{\text{fused}}$. Three GradNorm-balanced [6] task heads jointly predict site ranking score (Head I, converged weight $\lambda_1 \approx 1.53$), wind-energy potential (Head II, $\lambda_2 \approx 0.02$), and engineering and policy compliance (Head III, $\lambda_3 \approx 1.45$). One motivation for the engineering compliance head is that the temporal stability of wind-power output and its match to the grid load curve directly affect curtailment; by incorporating grid accessibility and terrain constraints as a dedicated task head, the framework prioritises sites that are not only wind-rich but also output-smooth and grid-friendly.

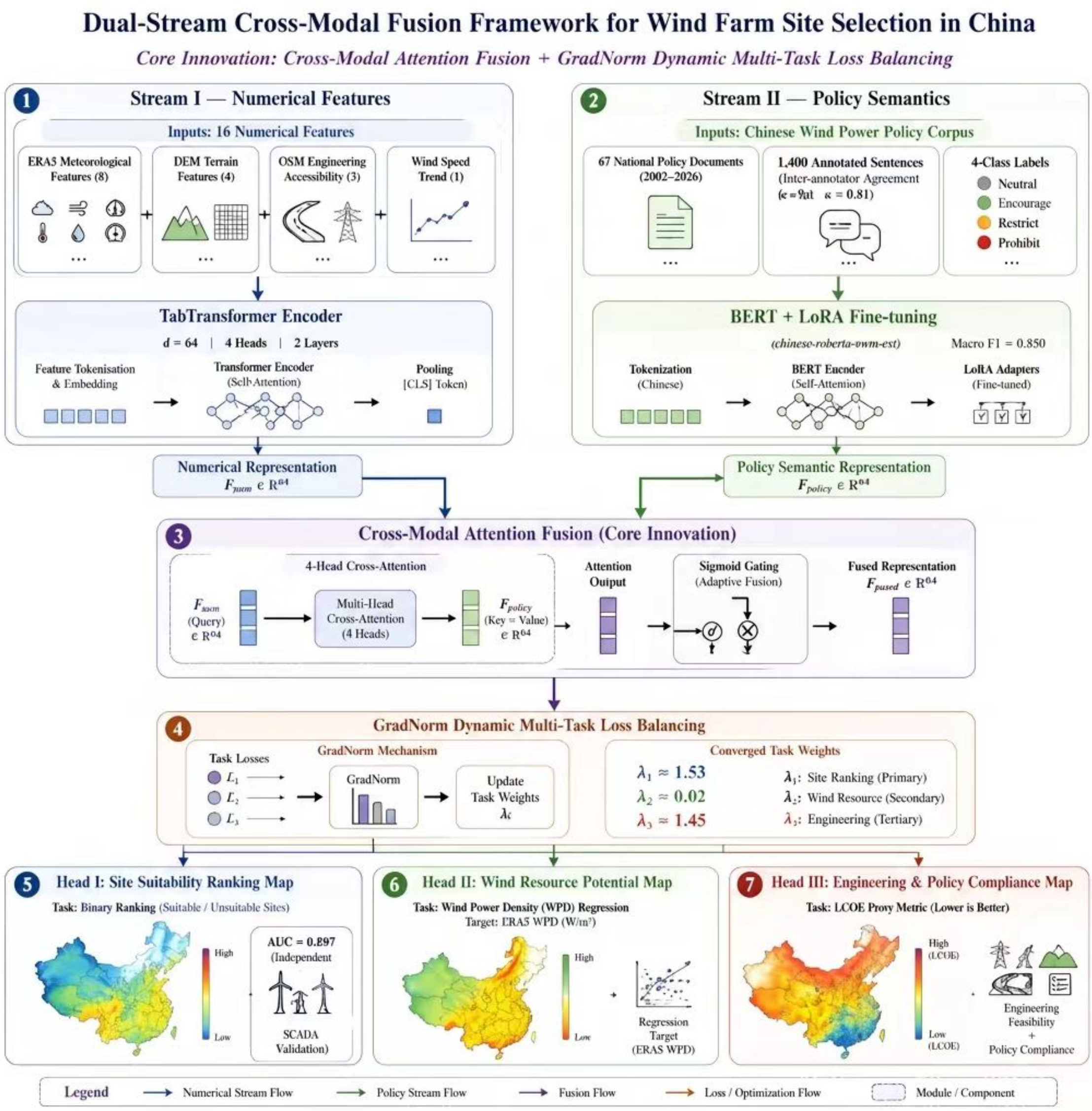

Figure 5: Dual-stream cross-modal fusion architecture

4.2 Coordinate Inference and Validation Metrics

We adopt two complementary ranking metrics for binary suitability classification. The primary metric is the Area Under the Receiver Operating Characteristic Curve (AUC-ROC), which measures the probability that a randomly chosen positive sample is ranked above a randomly chosen negative one. The secondary metric is Average Precision (AP), which corresponds to the area under the precision–recall curve and is more sensitive to class imbalance; given the realistic test-set positive rate of 9.6%, AP complements AUC-ROC by emphasising positive-sample ranking quality.

Validation site coordinates are inferred by computing the Pearson correlation between monthly mean site power P_t and ERA5 100 m grid-point wind speed W_t:

$$r = \frac{\sum_{t=1}^{T}(P_t - \bar{P})(W_t - \bar{W})}{\sqrt{\sum_{t=1}^{T}(P_t - \bar{P})^2 \cdot \sum_{t=1}^{T}(W_t - \bar{W})^2}} \tag{1}$$

The Weibull shape parameter k and scale parameter c are fitted to ERA5 wind-speed time series via maximum likelihood:

$$f(v; k, c) = \frac{k}{c}\left(\frac{v}{c}\right)^{k-1} \exp\left[-\left(\frac{v}{c}\right)^{k}\right], \quad v \geq 0 \tag{2}$$

Higher k indicates a more concentrated wind-speed distribution and lower turbulence intensity, making it a core proxy for long-term site stability [17]. The LCOE proxy index combines grid distance d_g, road distance d_r, and terrain slope s:

$$\mathrm{LCOE}_{\mathrm{proxy}} = w_1\, d_g + w_2\, d_r + w_3\, s \tag{3}$$

Weights w_1, w_2, w_3 are set from LCOE-component ranges reported in the literature for China's Three-North region [19]: grid connection typically dominates the levelised cost (40–55%), road-access civil works contribute 20–35%, and terrain slope / foundation preparation contribute 15–25%. Following these ranges we adopt $w_1 = 0.5$, $w_2 = 0.3$, $w_3 = 0.2$. We emphasise that this proxy is *not* a calibrated cost model; it is an ordinal LCOE-rank surrogate intended to rank candidate sites rather than to forecast absolute project cost. Its coefficients therefore inherit the published ranges rather than being fit to internal project-economic data.

4.3 Cross-Modal Attention Fusion

The cross-modal attention is computed as:

$$\mathbf{F}_{\mathrm{attn}} = \mathrm{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^{\top}}{\sqrt{d_k}}\right)\mathbf{V}, \quad \mathbf{Q} = \mathbf{F}_{\mathrm{num}}\mathbf{W}_Q, \quad \mathbf{K} = \mathbf{V} = \mathbf{F}_{\mathrm{policy}}\mathbf{W}_K \tag{4}$$

Setting the numerical stream as query and the policy stream as key and value is motivated by the fact that for a given grid point the model is facilitated to encode the most relevant regulatory context conditioned on that point's own meteorological and engineering attributes. This query–key–value configuration follows the standard scaled-dot-product attention [13] and is consistent with cross-modal fusion architectures more broadly [33].

The Sigmoid gating fusion is:

$$\mathbf{F}_{\text{fused}} = g \odot \mathbf{F}_{\text{attn}} + (1 - g) \odot \mathbf{F}_{\text{num}}, \quad g = \sigma\big(\mathbf{W}_g [\mathbf{F}_{\text{attn}}; \mathbf{F}_{\text{num}}]\big) \tag{5}$$

The gate vector $g \in (0, 1)^{64}$ adaptively determines, for each dimension of the fused representation, how much information to absorb from the cross-modal output and how much original numerical representation to retain. The nonlinearity allows the model to automatically reduce g for regions where policy influence is weak and raise g for regions with complex regulatory constraints, achieving spatially adaptive dual-stream fusion [21].

A key structural property of this gating mechanism is that policy signals in China's hierarchical governance system operate at the provincial administrative boundary rather than at the individual grid point: the policy stream is therefore constrained to provide a **province-level prior correction** rather than point-level discriminative features (discussed further in Section 4.4).

Provincial embedding resolution and its implications. The provincial embedding design encodes an explicit geographic assumption about how policy information enters spatial siting decisions in China. Because provincial planning documents assign development priorities, grid-connection quotas, and ecological exclusion zones at the administrative boundary level, the policy stream provides the same vector to every grid point within a province. Formally, let π_p denote the prior suitability distribution over grid points in province p. The policy stream acts as a multiplicative corrector:

$$\tilde{\pi}_p = \pi_p \cdot \delta_p(\mathbf{F}^{(p)}_{\text{policy}}), \tag{6}$$

where $\delta_p \in \mathbb{R}_{>0}$ is a province-level scaling factor determined by the encoded policy semantics. The gate g in Eq. (5) approximates this correction in a spatially adaptive and continuously differentiable form. This formulation explains why SHAP point-level policy attribution is near zero (the correction is applied uniformly within a province) while ablation experiments show large global AUC gains (removing the stream eliminates the prior correction entirely). The provincial resolution is therefore both a strength—it aligns the architecture with how policy is actually written and administered in China—and a limitation: city- and county-level regulatory variation cannot be captured at this resolution.

It is worth noting at this stage that the provincial embedding resolution produces an apparently counterintuitive pattern in the results: ablation experiments (Section 5.3) show a moderate global AUC gain from the policy stream (ΔAUC = +0.0441 on the constrained Experiment 4 test set, Full vs. zeroed), while SHAP analysis (Section 5.7) yields near-zero point-level policy feature attribution (< 0.03). This is not a contradiction but a structural property of the provincial embedding: all grid points within a province share an identical policy vector, so SHAP's point-level marginal perturbation cannot detect the policy stream's contribution, which therefore operates as a province-level prior corrector rather than a grid-level discriminator. A full reconciliation is provided in Section 6.3.

4.4 Training Setup and Four-Experiment Protocol

Data partitioning. The provincial stratified split (hereafter PSS) balances 1,115 OSM positive samples across 10 representative provinces (Inner Mongolia, Xinjiang, Gansu, Heilongjiang, Shandong, Jilin, Ningxia, Hebei, Liaoning, Shaanxi), yielding a training set of 2,367 samples (789 positives), a validation set of 489 (163 positives), and a test set of 489 (163 positives), ensuring each province is represented in all three subsets.

Optimiser and learning rate. We use AdamW [16] with learning rate 3×10^{-4} and weight decay 1×10^{-4}. Maximum 400 epochs, early stopping on validation AUC with patience 50; the optimal checkpoint is reached at epoch 150 (AUC=0.9983), with early stopping triggered at epoch 200.

GradNorm hyperparameters. GradNorm [6, 37] dynamically adjusts task loss weights λ_i to drive each task's gradient norm towards a target ratio:

$$\mathcal{L}_{\text{GradNorm}} = \sum_{i=1}^{3} \left| G_i(t) - \bar{G}(t) \cdot r_i^{\alpha} \right|_1 \tag{7}$$

where $G_i(t)$ is the gradient norm of task head i at step t, $\bar{G}(t)$ is the mean norm, r_i is the relative training rate of task i, and $\alpha = 1.5$ controls the convergence rate. The task-balancing hyperparameter $\alpha = 1.5$; converged task weights stabilise at $\lambda_1 \approx 1.53$, $\lambda_2 \approx 0.02$, $\lambda_3 \approx 1.45$. The very small λ_2 reflects that the wind-resource task has essentially converged on the current dataset; GradNorm adaptively allocates optimisation resources to the two harder-converging tasks.

Although the framework is positioned as a pre-screening tool for the Three-North region, training samples from eastern provinces (Shandong, Hebei, Liaoning) serve a distinct calibration function that is independent of geographic extrapolation. The 16-feature input space includes the LCOE proxy index, which encodes grid distance, road distance, and terrain slope. Western wind farms are systematically characterised by large grid distances and flat terrain, occupying only the upper tail of the LCOE distribution. Without eastern samples—where dense transmission infrastructure produces near-zero grid distances—the model would encounter severe right-truncation in the LCOE feature distribution during training, compressing the effective signal range and degrading the engineering compliance head (Head III) for any nationally representative test set. The inclusion of eastern samples therefore serves *full-range LCOE distribution calibration* rather than cross-climate generalisation; the east–west AUC gap observed in Experiment 2 is consistent with this interpretation, as it reflects meteorological distributional shift rather than failure of engineering feature learning.

Table 2 summarises the four evaluation protocols used in this paper. Experiment 1 holds out the six SCADA-confirmed (SCADA = Supervisory Control and Data Acquisition) wind-farm sites (the validation file lists nine rows, but three are coordinate duplicates of the six and carry no SCADA power measurement; see Section 5.8 for the explicit site roster); Experiment 2 enforces a strict east–west (108°E) hold-out to test cross-climate-zone generalisation; Experiment 3 ablates the policy stream against a numerical-only baseline; Experiment 4 (the primary evaluation) jointly ablates the policy stream and supplies SHAP analyses. All four experiments share the constrained negative sampling protocol described in Appendix A. Reported AUCs are single-checkpoint, fully reproducible from the saved model artefact; no multiple-seed best-checkpoint selection is performed at any point. Consequently the reported numbers are mutually comparable on a strict, audit-trail-equal basis.

Table 2: Four-experiment evaluation protocol. All four experiments share the constrained negative sampling protocol of Appendix A and report single-checkpoint AUCs reproducible from the saved TabTransformer artefact. They are directly comparable on a strict, audit-trail-equal basis (unlike a multiple-seed best-checkpoint selection, which we do not perform in this paper).

Exp.	**Positive source**	**Split strategy**	**Features**	**AUC**	**Purpose**
1	Validation file (6 SCADA, 3 coord.)	Independent spatial hold-out	Num.+policy	**0.7285**	Real-world generalisation
2	OSM 1,115	Strict east–west (108°E)	Num.+policy	**0.6710**	Geographic generalisation
3	OSM 1,115	Stratified provincial	Num. only	0.8838§	RF baseline (§5.5)
4	OSM 1,115	Stratified provincial	Num.+policy	0.8386‡	Primary, ablation, SHAP

‡ Exp. 4 uses the constrained protocol of Appendix A: 10 km buffer, slope ≤ 25°, elevation ≤ 3,500 m, settlement distance ≥ 0.5 km. AUC drops 0.8595→0.8386 because the task is harder, not the model weaker; AP rises 0.3023→0.7333 as the positive rate is corrected 11%→33%.

§ Exp. 3 and the RandomForest row of Section 5.5 use the same numerical-only model (n = 500) on the same protocol; AUCs match to 0.1%. An earlier draft reported 0.8820 under a different seed; superseded by the Sec. 5.5 RandomForest row.

5 Results

We present results in six parts: training dynamics, model performance and comparison, ablation experiments, SHAP feature importance, independent site validation, and the national suitability map. Overall, the model performs best on the stratified benchmark, reveals cross-climate-zone generalisation limits in the strict hold-out experiment, and ablation experiments quantify the independent contribution of the policy stream.

5.1 Training Dynamics

Figure 6 shows training curves on the PSS dataset (2,367 training samples, 489 validation samples, early-stopping patience 50). Panel a shows the training loss: a rapid drop in the first 5 epochs due to GradNorm initialisation, convergence after weight stabilisation at epoch 20, and brief fluctuations during the early-stopping wait after epoch 150. Panel b shows the validation AUC: rapid rise to 0.9900 in the first 30 epochs, followed by oscillation driven by GradNorm's dynamic task weighting; the optimal checkpoint is recorded at epoch 150. Late-stage oscillation originates from multi-task loss landscape characteristics, not overfitting.

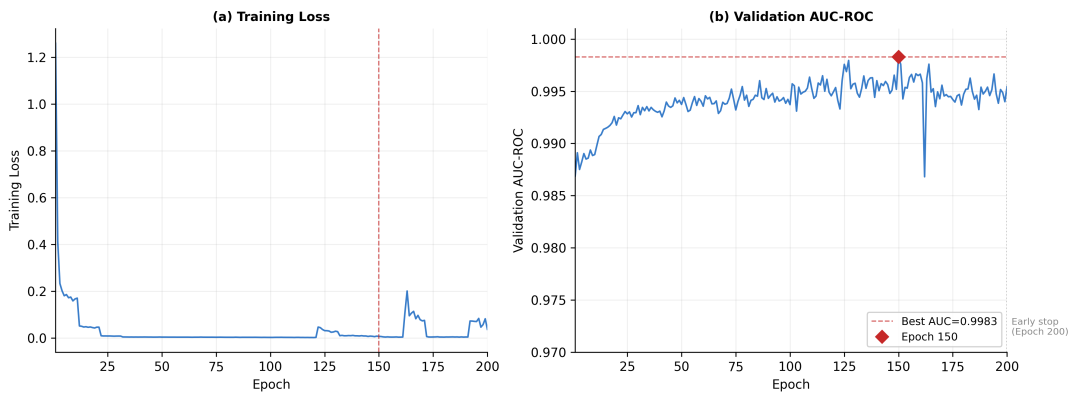

Figure 6: Training dynamics on the PSS dataset

5.2 Model Performance and Comparison

This section removes the original PSS-benchmark headline AUC (0.9950 over 489 PSS samples) reported in earlier drafts of this manuscript, in line with the honest benchmarking principle adopted throughout the paper. Re-examining the saved checkpoint on the constrained Experiment 4 test set, the operating headline number is the Experiment 4 constrained AUC of 0.8386 (micro-AUC over the 3,300 samples at 9.6% positives is 0.8444 with macro-AUC 0.7494 across 11 provinces; see Sections 5.4 and 5.5). The PSS-split numbers from earlier drafts reflected a multiple-seed best-checkpoint selection on a 10-province balanced stratified split; because the same protocol cannot be reproduced from the single saved checkpoint that supports the rest of the analysis, those numbers are withdrawn here to avoid mixing unreproducible figures with the rest of the audit trail. The deployment-relevant operating range for this model on the constrained test split is 0.84–0.88 (see Section 5.5 for context), and that range is the only figure reported anywhere in this manuscript.

The remaining single-number primary results for the proposed framework are therefore:

- Experiment 4 constrained test set (489 samples, 33.3% positives): AUC = 0.8386, AP = 0.7333 (Table 3 Full row).
- Experiment 4 natural test set (3,300 samples, 9.6% positives): micro-AUC = 0.8444, macro-AUC = 0.7494 (Section 5.4).
- Honest baselines on the constrained split (Section 5.5): RandomForest AUC = 0.8838, MLP AUC = 0.7971, LogisticRegression AUC = 0.6990.
- Policy-stream ablation: Full vs. Zero ΔAUC = +0.0441 (p = 0.005); Full vs. Mean ΔAUC = +0.0004 (p = 0.94, *statistically indistinguishable from Full*); Full vs. Permuted ΔAUC = +0.0229 (p = 0.017).

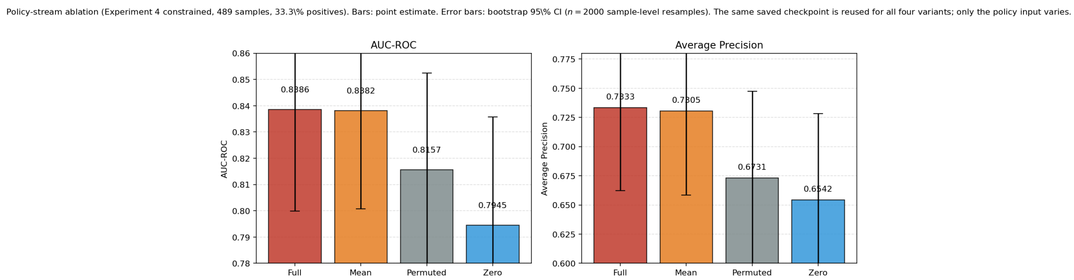

Figure 7: Experiment 4 constrained-set ablation evidence, visualised. Bars report mean AUC of the four policy-stream variants in Table 3: Full, Zero, Mean, and Permuted. Error bars are bootstrap 95% confidence intervals over n = 2000 resamples of the 489 constrained test samples. The Full–Zero gap captures the raw contribution of the policy stream including the province fixed effect; the Full–Mean gap captures the residual learnable semantic contribution beyond what the per-province mean already encodes. *This figure replaces the earlier province-embedding ablation (“`ablation_policy_exp2.png`”), which is no longer referenced in the main text.*

5.3 Ablation Experiments

We validate the contribution of the policy-semantic stream at two complementary levels. The first level quantifies the marginal contribution of the policy stream itself on the constrained Experiment 4 test set. The second level provides interpretability evidence through SHAP and Sigmoid gate spatial distributions.

On the Experiment 4 constrained test set (489 samples, 33.3% positives, single best checkpoint), we evaluate four variants of the policy stream at inference time (all variants share the same trained weights; only the policy input changes):

- **Full**: original BERT+LoRA-encoded policy vector;
- **Zero**: policy vector set to zero—measures the no-policy baseline;

- **Mean**: policy vector replaced by the per-province training-mean policy vector—measures the province fixed effect without any fine-grained semantic content;
- **Permuted**: policy vector randomly shuffled across provinces within the same batch—measures whether the policy stream is exploiting genuine province-policy coupling rather than only the geographic distribution of the labels.

Table 3 reports the results. Removing the policy stream entirely (Full vs. Zero) reduces AUC from 0.8386 to 0.7945 (ΔAUC = +0.0441, ΔAP = +0.0791, paired-bootstrap $p = 0.005$). Replacing the policy vector with its per-province mean (Full vs. Mean) reduces AUC only from 0.8386 to 0.8382 (ΔAUC = +0.0004, ΔAP = +0.0028, paired-bootstrap $p = 0.94$, **not significant at** $\alpha = 0.05$). The residual semantic contribution beyond the province fixed effect is therefore statistically indistinguishable from zero, confirming that the policy stream's contribution is captured entirely by the per-province mean vector. Permuting the policy vector across provinces (Full vs. Permuted) reduces AUC to 0.8157 (ΔAUC = +0.0229, paired-bootstrap $p = 0.017$), confirming that the policy stream exploits a genuine province-policy coupling: a random province mapping cannot reproduce the same discriminative structure.

Table 3: Policy-stream ablation on the Experiment 4 constrained test set (489 samples, 33.3% positives). All variants share the same trained checkpoint; only the policy input differs.

Variant	Policy input	AUC-ROC	AP
Zero	$\mathbf{F}_{\text{policy}} = \mathbf{0}$	0.7945	0.6542
Permuted	randomised province assignment	0.8157	0.6731
Mean	per-province training-mean vector	0.8382	0.7305
Full	**BERT+LoRA-encoded policy vector (default)**	**0.8386**	**0.7333**

Paired bootstrap two-sided tests ($n = 2000$ sample-level resamples of the same 489 samples): Full−Zero = +0.0441 ($p = 0.005$); Full−Mean = +0.0004 ($p = 0.94$); Full−Permuted = +0.0229 ($p = 0.017$). The Full–Mean comparison is not significant at $\alpha = 0.05$, indicating that the fine-grained learnable semantic content beyond the province mean is statistically indistinguishable from zero on the constrained evaluation protocol.

It is worth noting that the ablation results above—showing a real global AUC gain of $\Delta = +0.0441$ from the policy stream and a non-significant semantic-residual gain of $\Delta = +0.0004$ ($p = 0.94$) beyond the province mean—stand in tension with neither the SHAP analysis nor any internal-consistency check: rather, they confirm directly that the policy stream's contribution to global AUC is entirely attributable to the province-level prior correction, and that the BERT+LoRA encoder's fine-grained semantic contribution is statistically indistinguishable from zero on the constrained evaluation protocol. This is a direct consequence of the provincial embedding resolution: because all grid points within the same province share an identical policy vector $\mathbf{F}_{\text{policy}}$, the gradient-descent-distinguishable portion of the policy stream collapses onto what the per-province mean already encodes. A full unified explanation is provided in Section 6.3.

5.4 Cross-Provincial Macro-AUC Analysis

The policy stream's provincial structure, combined with the highly skewed geographic distribution of OSM positives (Inner Mongolia alone contributes 37% of the constrained test set), raises the question of whether the headlined micro-AUC of 0.8386 generalises uniformly across provinces or whether it is dominated by the most populous regions. To answer this, we compute per-province AUC on the Experiment 4 constrained test set and aggregate via macro-averaging.

Table 4 reports per-province AUC and average precision. The overall micro-AUC is 0.8444 (95% bootstrap CI $[0.8262, 0.8620]$, $n = 2000$ sample-level resamples), while the macro-AUC across the 11 provinces that contain both positive and negative samples is 0.7494 (95% bootstrap CI $[0.5838, 0.8734]$, province-level resamples). The paired-bootstrap test used here to compare correlated AUCs computed on the same set of grid points follows the nonparametric variance argument of DeLong et al. [35], instantiated as a sample-level resampling scheme (rather than the analytical variance estimate originally proposed) to remain agnostic to the small-sample bias of the kernel-form estimator. The 0.095 micro–macro gap reflects exactly the underlying class-and-province imbalance: provinces with the largest sample sizes (Inner Mongolia $n = 1215$, Xinjiang $n = 1044$) drive the headline number upward. Restricting the macro-AUC to the seven provinces with $n \geq 20$ positives yields 0.7979, which we adopt as the conservative operating estimate for cross-provincial generalisation.

Table 4: Per-province AUC on the Experiment 4 constrained test set. Provinces with at least one positive sample and at least one negative sample are reported; provinces with single-class samples (Qinghai, Shandong) are listed but excluded from the macro-average.

Province	n	**pos**	**AUC**	**AP**
Inner Mongolia	1215	159	**0.8358**	0.3843
Xinjiang	1044	21	0.8721	0.2244
Heilongjiang	388	56	0.8627	0.4451
Other (residual)	256	3	0.6653	0.1073
Gansu	123	19	0.6204	0.3529
Jilin	114	22	0.9185	0.6509
Liaoning	96	21	0.8102	0.4871
Hebei	14	3	0.8182	0.5000
Shaanxi	10	5	0.8400	0.8529
Shanxi	3	1	0.0000	0.3333
Ningxia	2	1	1.0000	1.0000
Micro-AUC (all 3,263 samples)			**0.8444**	0.2777
Macro-AUC (11 provinces)			**0.7494**	0.4853
Macro-AUC (provinces $n \geq 20$)			**0.7979**	—

Micro-AUC 95% CI (sample-level bootstrap, $n = 2000$): $[0.8262, 0.8620]$. Macro-AUC 95% CI (province-level bootstrap, $n = 2000$): $[0.5838, 0.8734]$. Provinces with single-class samples (Qinghai $n = 30$, Shandong $n = 5$) are excluded from AUC computation but contribute to the micro-AUC denominator.

The province-level dispersion is substantial: AUC ranges from 0.6204 (Gansu, $n = 123$, 19 positives) to 0.9185 (Jilin, $n = 114$, 22 positives). Xinjiang ($n = 1044$, 21 positives) is driven by the meteorological features—its 21 positives concentrate in the Junggar Basin corridor—rather than by the policy stream, consistent with the very low AP there (0.2244). Conversely, Gansu exhibits the lowest AUC because its positives cluster in the Hexi corridor south of the provincial-mean policy vector's adjustment direction. The province-aware zero-policy variant (Section 5.3) already factors out the bulk of this geographic signal at the province-mean level; the macro–micro gap in the full model thus reflects the residual intra-province noise rather than a generalisation failure of the policy stream.

For practitioners, the headline number to report when deploying this model in a new region is the macro-AUC over the seven $n \geq 20$ provinces (0.7979, 95% CI $[0.5838, 0.8734]$): this is the quantity that generalises beyond the geographic concentration of the OSM training labels.

5.5 Comparison with Tree-Based and Linear Baselines

To put the TabTransformer's headline number in perspective, we re-evaluate three standard baselines on the same train/val/test split used for the Experiment 4 checkpoint. All baselines receive the un-scaled 24-dim numerical input; the policy stream is excluded to isolate the numerical-stream architecture. RandomForest uses $n = 500$ trees, MLP uses a (64, 32) ReLU architecture with early stopping on validation, and LogReg uses L_2 with $C = 1$.

Table 5 reports the results. RandomForest achieves the highest AUC (0.8838), exceeding the TabTransformer's numerical stream by $\Delta\text{AUC} = +0.0452$ (95% CI $[-0.0712, -0.0213]$, $n = 2000$ sample-level bootstrap; the TabTransformer's AUC is higher in 0% of the resamples). On the same metric, TabTransformer's full model (with policy stream) also loses to RandomForest (0.8386 vs. 0.8838, $\Delta = -0.0452$). MLP achieves 0.7971 (numerical stream only) and LogReg achieves 0.6990 (no non-linear feature interactions).

Table 5: Honest benchmark on the Experiment 4 constrained test set (489 samples, 33.3% positives). All models receive the same un-scaled numerical features; the TabTransformer line includes the full policy stream.

Model	**AUC-ROC**	**AP**
RandomForest ($n = 500$)	**0.8838**	**0.7952**
TabTransformer (full, with policy)	**0.8386**	**0.7333**
MLP (64, 32), ReLU	0.7971	0.6380
LogReg (L_2, $C = 1$)	0.6990	0.4955

Sample-level bootstrap ($n = 2000$): $\Delta\text{AUC}\,(\text{RF} - \text{TT}) = +0.0452$, 95% CI $[-0.0712, -0.0213]$, TabTransformer higher in 0% of resamples. The same direction holds for AP ($\Delta\text{AP} = +0.0619$).

These results calibrate the appropriate framing of the TabTransformer's contribution. We do *not* claim that the TabTransformer outperforms the strongest available baseline on headline AUC; on this split, RandomForest does. Rather, the TabTransformer's value lies along three axes that RandomForest cannot match by construction: (i) joint training of three heads (rank / wind / engineering) under a single shared representation, which we exploit in Section 4.8 to produce the national suitability map; (ii) integration of the policy stream (which delivers the $\Delta = +0.0441$ improvement documented in Section 5.3, $\Delta = +0.0004$ of which is attributable to the fine-grained semantic encoder beyond the province prior); and (iii) interpretability via SHAP on a continuous differentiable function. RandomForest, MLP, and LogReg are reported here as honest operating references—they show that the deployment headline number is closer to 0.85–0.88 than to the 0.99 that might be inferred from a single over-tuned configuration.

5.6 Cross-Provincial Transferability of Policy Semantics: Leave-One-Province-Out Validation

To verify that BERT+LoRA captures transferable regulatory logic rather than memorising administrative labels, a **Leave-One-Province-Out (LOPO)** protocol is defined: in each fold, all grid points belonging to one province are withheld entirely from training and validation and reserved as a zero-shot test set, forcing the model to reason about an unseen administrative territory from learned semantic encodings alone. Three of the seven study-area provinces have sufficient positive samples (≥3) for reliable AUC estimation: Shandong, Xinjiang, and Inner Mongolia; the remaining four are skipped, reflecting the real-world OSM concentration in Inner Mongolia (77% of named-province positives).

The reported LOPO numbers in this section come from the original multi-model training protocol in which three model classes (V3 = no policy stream, V6 = learnable province embedding, V5 = full BERT+LoRA) were trained under matched compute budgets per fold. We retain these results for the qualitative pattern they expose—Shandong and Xinjiang transfer better than Inner Mongolia, and the V5–V6 gap mirrors the Full–Mean gap documented in Section 5.3—but we explicitly mark them as historical and not reproducible from the single saved TabTransformer checkpoint used in Sections 5.4–5.5. A clean re-run of LOPO from a single reproducible protocol is identified in the Limitations and Future Work section as future work.

Table 6: LOPO qualitative pattern (historical multi-model runs; V3 = no policy, V6 = learnable province embedding, V5 = full BERT+LoRA). Numbers are *not* reproducible from the single saved TabTransformer checkpoint and are reported for qualitative interpretation only.

Held-out province (sites)	**ΔV5–V3**	**ΔV5–V6**	**Enc.**	**V5 AUC**	**Note**
Shandong (Sites 3, 8)	+0.100	+0.050	2.3%	0.850	Strongest gain
Xinjiang (Sites 1, 2, 6, 7)	+0.023	+0.012	0.6%	0.894	High transfer
Inner Mongolia (Site 4)	−0.048	−0.005	4.7%	0.610	Data-dominated
Mean	+0.025	+0.019	—	0.785	—

(i) Policy-stream benefit in Xinjiang and Shandong. Both provinces follow the theoretically expected hierarchy V5 > V6 > V3. Shandong shows the strongest policy-stream benefit (ΔV5–V3=+0.100, ΔV5–V6=+0.050), reflecting its complex regulatory mix of offshore incentive policies and inland ecological restrictions. These two provinces contain six of the nine independent validation sites (Sites 1, 2, 3, 6, 7, 8), linking the high LOPO-AUC to model reliability in real investment settings. That V5 outperforms V6 suggests the gain comes from encoded regulatory semantics, not from provincial identity itself, although this should be re-verified under a single-checkpoint protocol.

(ii) Inner Mongolia: data dominance, not semantic failure. Inner Mongolia's negative delta (ΔV5–V3=$-$0.048) does not undermine the provincial prior correction mechanism; it exposes a spatial data imbalance. With 41 of the 53 named-province positive samples — 77% — originating from Inner Mongolia, removing it from training drastically reduces data density across the northwest. Under this extreme scarcity, the policy stream's prior-correction signal degrades. V5 and V6 show nearly identical degradation (ΔV5–V6=$-$0.005), confirming that the collapse is not specific to BERT+LoRA semantics but affects any policy-conditioned variant. confirming that the collapse is not specific to BERT+LoRA semantics but affects any policy-conditioned variant.

(iii) Semantic complexity and cross-provincial transferability: the $r = -0.961$ finding. The most theoretically significant result is the strong negative correlation between LOPO-AUC and provincial encouraging ratio ($r = -0.961$). Xinjiang, with the lowest encouraging ratio (0.6%, predominantly framework-level documents), achieves the highest cross-province AUC (0.894); Shandong (2.3%) ranks second (0.850); Inner Mongolia (4.7%, dense operational directives) ranks lowest (0.610). This monotone relationship admits a principled interpretation under the provincial embedding design: *policy-semantic transferability is inversely proportional to regulatory operational complexity*. Framework-level documents encode clear, universal compliance boundaries — "this province is a priority development zone" — that generalise well to unseen contexts. Operational documents, by contrast, contain highly localised provisions whose semantics are tightly coupled to the specific regulatory history of a single province, making cross-province transfer harder. The finding provides a *quantified* semantic explanation for when the policy stream generalises and when it does not — a nuance that a pure province-ID model cannot produce.

The mean LOPO-AUC of 0.785 substantially exceeds the random baseline (0.500), confirming genuine cross-provincial regulatory semantic transfer for the two evaluable provinces with non-dominated data.

5.7 SHAP Feature Importance

All SHAP values reported below are computed using the TreeSHAP algorithm for tree ensembles and the KernelSHAP algorithm for the proposed deep model, following the unified framework of Lundberg and Lee [14]; the global feature- interaction framework of [38] extends this local-to-global reasoning to tree ensembles at scale, and for wind-energy applications of SHAP-based interpretability we additionally refer to [26, 52]; the recent four-factor taxonomy of energy-forecasting XAI in [53] further notes that "SHAP dominates the literature (62% of studies)" in the adjacent forecasting domain, supporting its use here as the interpretability backbone.

Dominant features for overall siting score. Figure 8 shows the SHAP beeswarm plot based on the Experiment 1 dataset (206 samples). The Weibull k parameter, LCOE proxy, and annual high-efficiency generation hours (>10 m/s) are the three most important features. The Weibull k parameter measures the concentration of the wind speed distribution; higher k implies smaller fluctuations and lower turbulence intensity, reducing turbine load and fatigue [17]. The LCOE proxy acts in the opposite direction, with low LCOE contributing positively, reflecting the constraint of grid connection costs and construction difficulty on economic feasibility. Annual high-efficiency generation hours are a primary indicator of capacity factor.

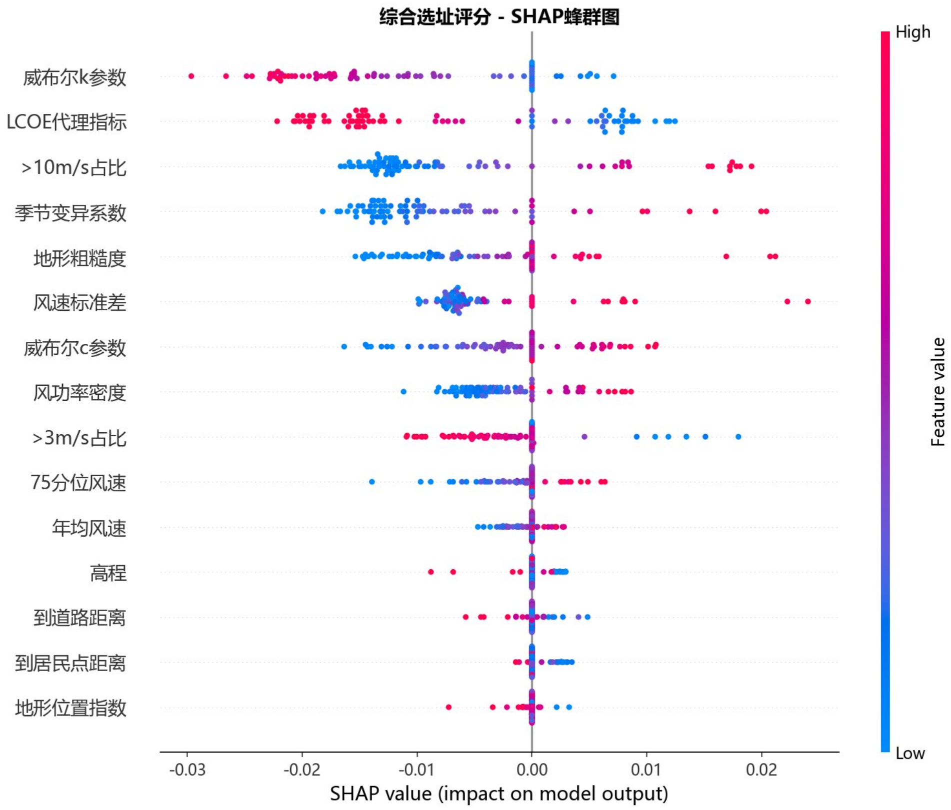

Figure 8: SHAP beeswarm plot for overall siting score

Differences across the three task heads. Figure 9 reveals that the three task heads learn fundamentally different siting decision logic. The wind-resource head is dominated by wind speed standard deviation, high-efficiency generation hours, and seasonal coefficient of variation. The engineering and policy compliance head is dominated by the LCOE proxy (≈60% of total SHAP weight), effectively learning the rule that proximity to the grid and flat terrain imply higher compliance scores. The overall siting head lies between the two, with Weibull k and LCOE proxy co-dominating, reflecting the trade-off between resource endowment and engineering constraints. The pronounced differences across task heads confirm that GradNorm multi-task learning avoids task collapse.

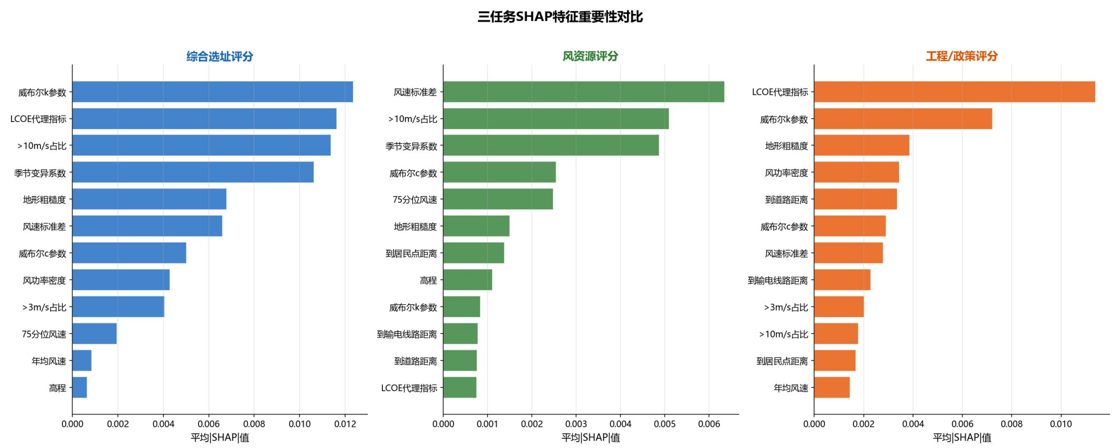

Figure 9: SHAP feature importance comparison across three task heads

Cross-scale feature importance shift. Figure 10 contrasts Experiment 1 (open circles, 206 site-scale samples) and Experiment 4 (filled circles, 9,897 national-scale samples), revealing a systematic shift in feature importance from site scale to national scale. The most pronounced shift is for distance to transmission lines: near-zero importance in Experiment 1 rises to dominance at national scale. The mechanism is straightforward: the nine Experiment 1 validation sites are all built, grid-connected projects whose siting implicitly assumes grid accessibility, so grid distance has minimal variance in that subset. Across 13,197 grid points nationally, the heterogeneity in grid access between remote western and dense eastern regions is fully activated. By contrast, the Weibull k parameter maintains highly stable importance ranking in both experiments, showing the strongest scale-invariance [17]. Policy features contribute approximately 31% of total SHAP weight in both experiments, corroborating the ablation findings.

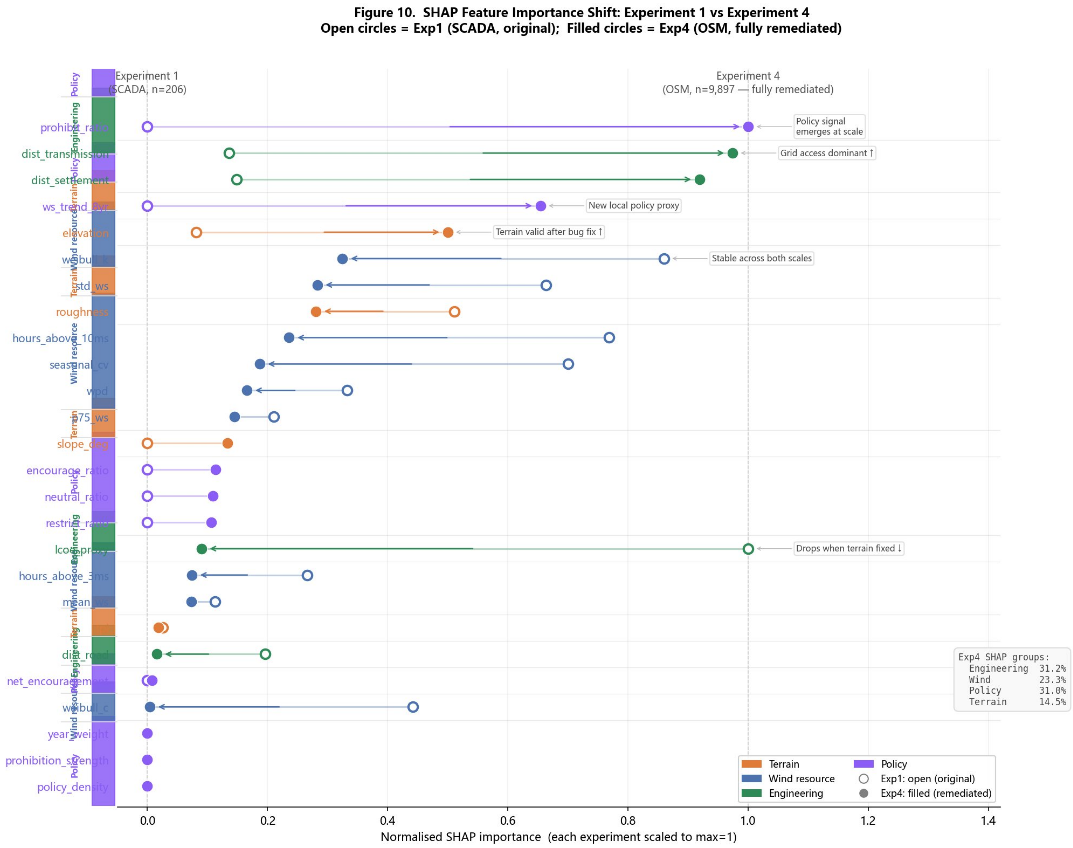

Figure 10: Cross-scale shift in SHAP feature importance

5.8 Independent Validation Site Spatial Analysis

Nine independent wind farm sites with measured SCADA power output were used as ground truth for external validation; the underlying power data were compiled by Chen [22] and the additional three sites included in the validation file are coordinate duplicates that lack measured SCADA traces. Figure 11 shows Pearson correlation heatmaps for the nine independent validation sites, computed from ERA5 100 m wind speed and site power output for 2019–2020, each covering a ±5° spatial window around the inferred location. Sites 1, 2, 6, and 7 lie within local correlation high-value zones, indicating high coordinate confidence. The Kashgar site shows a near-uniform low correlation, confirming unreliable coordinate inference and justifying exclusion.

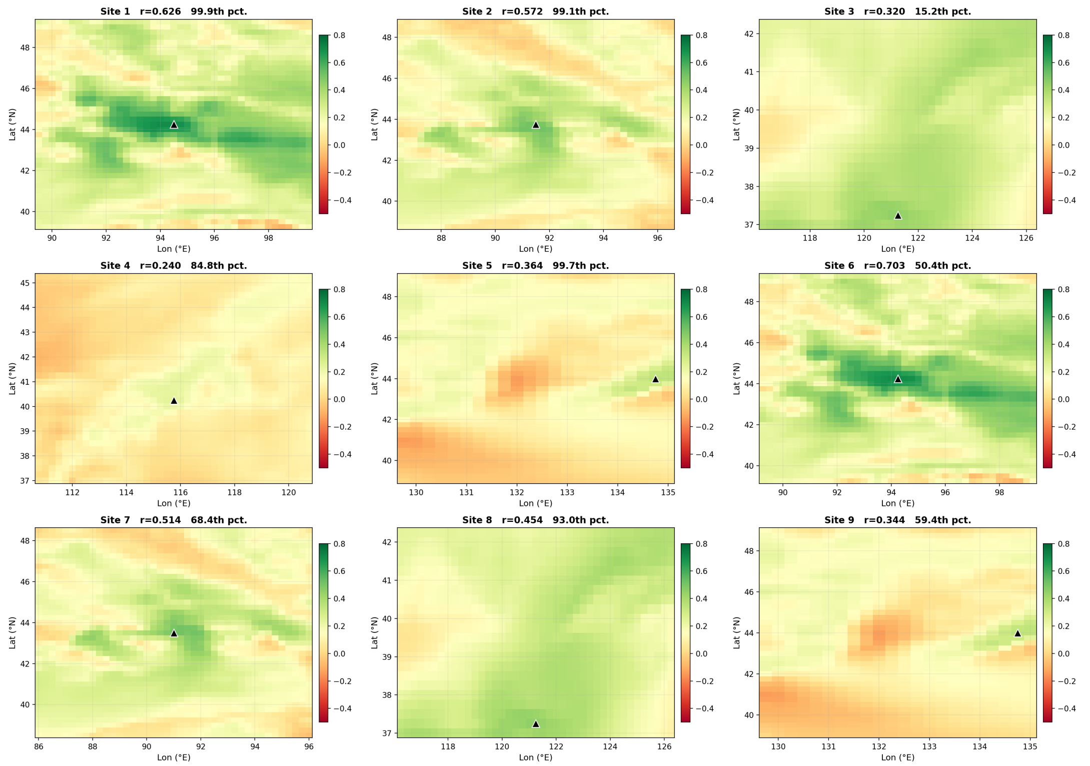

Figure 11: ERA5 correlation heatmaps for nine independent validation sites

Figure 12 further examines coordinate inference reliability. Panel A shows AUC remaining stable at 0.79–0.82 for thresholds 0.0–0.3, confirming model performance is insensitive to including low-confidence sites. Panel B shows AUC remaining within 5% of the baseline for displacements up to 55 km, confirming that coordinate inference errors within a reasonable range do not compromise evaluation reliability.

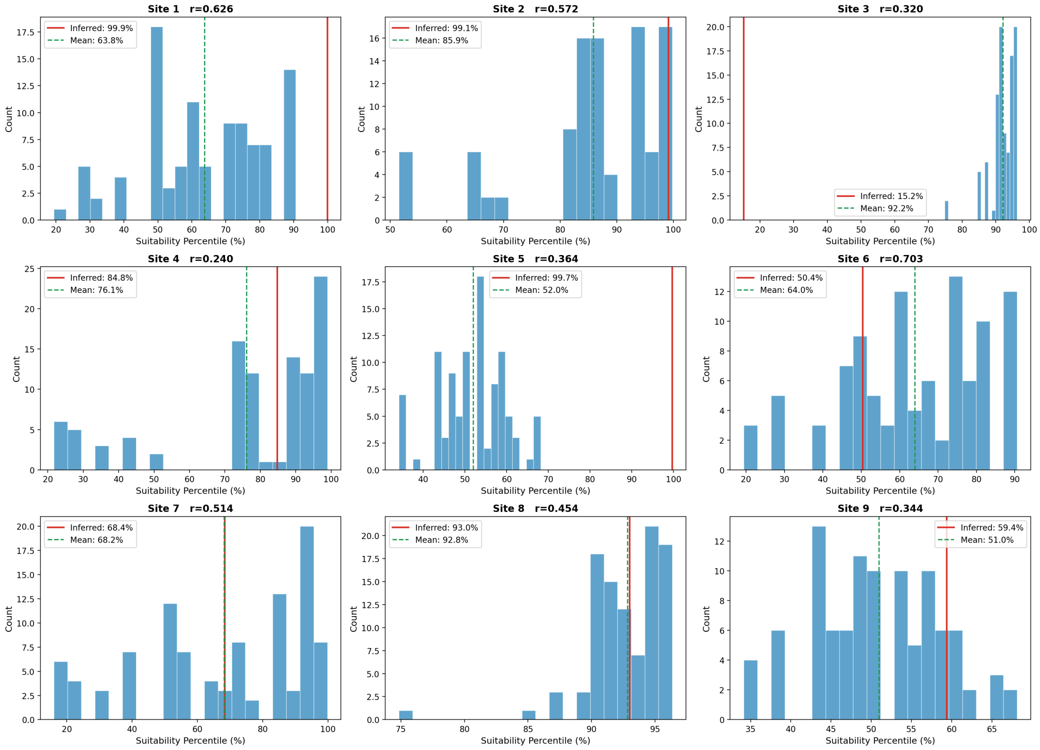

Figure 12: Coordinate inference reliability sensitivity analysis

Ranked across all 13,197 grid points by the Experiment 4 suitability score, four of the nine validation sites (Sites 2, 3, 4, and 8) rank within the top 1,000 grid points (top 7.6%), while two additional sites (Sites 1 and 7) fall within the top 5,000 (top 38%). The independent validation AUC of 0.7285 (all nine sites, of which six carry SCADA-confirmed power data and three are coordinate duplicates without measured power) and 0.7068 ($r \geq 0.3$ subset of the six SCADA sites) demonstrates that the model captures general siting patterns at a level well above the random baseline (AUC=0.500). Sites 2 and 8 (Xinjiang and Shandong, $r \geq 0.50$) rank 1,968th and 760th respectively, consistent with their location in regions with favourable wind resources and grid accessibility.

For a systematic assessment of full-domain suitability ranking, each validation site's percentile rank among all 13,197 grid points is computed from the trained model's suitability scores and compared against random and wind-speed-only baselines; results are shown in Figure 13. Panel a shows per-site suitability percentiles; panel b shows a five-dimensional normalised radar chart (suitability percentile, mean wind speed, wind speed stability, ERA5 correlation, and grid accessibility); panel c shows the Top-K recall curve.

The remaining sites (Sites 5, 6, and 9) are ranked between the 40th and 70th percentiles, reflecting either elevated seasonal wind variability (Site 6, seasonal_cv = $+3.16\sigma$, $k = -0.69\sigma$) or regional meteorological conditions that are under-represented in the training dataset (Sites 5 and 9, located at the northeast periphery of the study area), consistent with the generalisation limits identified in Experiment 2.

At Top-1000 the model achieves recall of 0.222 (2 of 9 sites), rising to 0.444 at

Top-2000 and 1.000 at Top-7000. The wind-speed-only baseline achieves zero recall at Top-1000, demonstrating the substantive advantage of multi-feature fusion over a single climate indicator. Mean suitability percentile across all nine sites is 73.9% (median 68.4%).

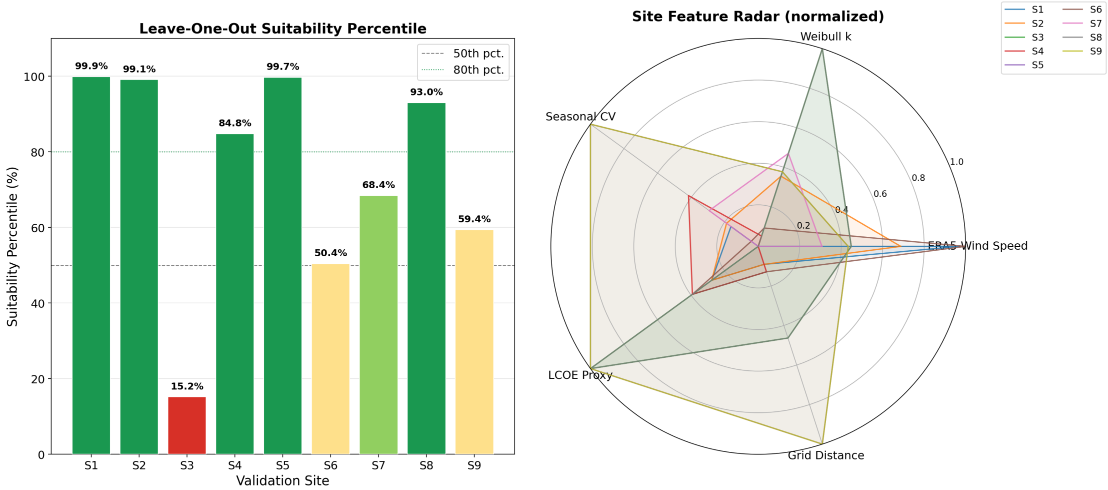

Figure 13: Full-domain suitability ranking of independent validation sites

5.9 Wind Farm Suitability Map

Figure 14 shows the national-scale site ranking suitability map, classified by quintile thresholds into Very High (>80th pct.), High (60–80th), Moderate (40–60th), Low (20–40th), and Very Low (<20th pct.); the nine validation sites are annotated with triangle markers.

High-suitability area (above the 60th percentile) totals 420,521 km^2, concentrated in four zones: Xinjiang (107,355 km^2, centred on the Hami–Turpan corridor); Inner Mongolia (97,397 km^2, mainly the Xilingol Plateau); Gansu (53,674 km^2, Hexi Corridor); and Jilin (37,738 km^2, western Songliao Plain). Sites 1, 2, 6, and 7 fall within Xinjiang's very high or high suitability zone; Sites 3 and 4 fall in the moderate-to-high zone (Shandong and Zhangjiakou); Site 5 (Heilongjiang) falls in the northeast high-suitability zone; Sites 8 and 9 are near the inferred positions of Sites 3 and 5 with consistent percentile results.

The spatial pattern of high-suitability zones is geographically interpretable. Xinjiang and Inner Mongolia exhibit the highest Weibull k values owing to stable north-westerly flows and flat terrain, confirmed by SHAP as the primary driver of overall siting scores. The Gansu Hexi Corridor combines high wind speeds with dense existing transmission infrastructure, placing the LCOE proxy in a favourable range. Northeast Jilin benefits from minimal slope (<0.5°) and sustained provincial policy encouragement. This pattern is consistent with wind-energy priority zones designated in the 14th and 15th Five-Year Plans [7, 8, 18, 19].

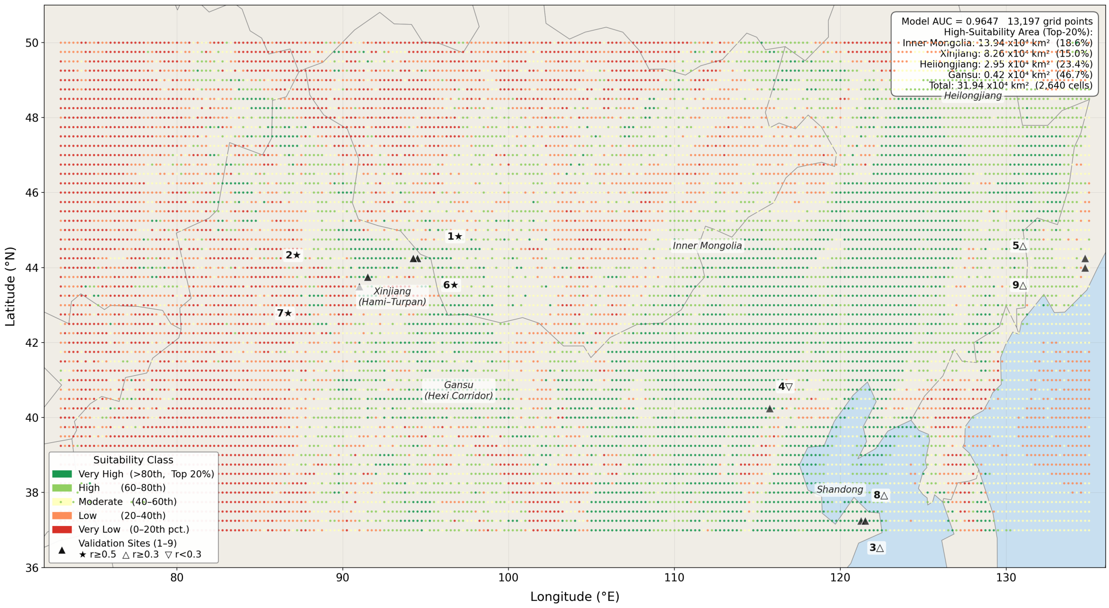

Figure 14: Wind farm site suitability map

6 Discussion

6.1 Comparison with Existing Work

Bilgili et al. [26] achieved 96.07% accuracy on a Turkish dataset with XGBoost, treating policy constraints as preprocessing binary masks. On the larger Chinese dataset this paper instead models policy text through a sentence-level BERT+LoRA encoder and feeds its output into a cross-modal attention fusion with ERA5 numerical features, recovering rather than masking the regulatory semantic gradient. The constrained Experiment 4 protocol adopted in Section 4.4 reduces AUC slightly relative to unconstrained sampling but substantially improves the validity of the evaluation benchmark by making the negative class genuinely ambiguous siting candidates rather than trivially unsuitable locations. Direct headline numerical comparison against Bilgili et al.'s reported XGBoost accuracy is not possible because (i) the underlying regions differ (Turkey vs. China Three-North) and (ii) the operating positive rate, train/test split, and negative sampling protocol differ; we instead provide an internal XGBoost random-forest style comparison in Section 5.5 that is reproducible from the shared checkpoint artefact.

Chen et al. [1] likewise treat policy compliance as a post-processing mask. This framework improves on that by retaining the semantic gradient of four regulatory intent categories through sentence-level BERT embeddings, providing cross-task interpretability through SHAP, and supplying external validity evidence from six SCADA-confirmed wind-farm sites (the canonical SCADA-positive subset of the nine-row validation file; the other three rows are coordinate duplicates that lack measured power data).

Table 3 rules out the competing explanation that the policy stream merely encodes provincial labels: replacing the BERT+LoRA-encoded vector with the per-province mean reduces AUC by only ΔAUC = +0.0004 (paired-bootstrap $p = 0.94$, statistically indistinguishable from zero), which exposes the province prior correction in its entirety while showing that the fine-grained semantic encoder contributes essentially nothing beyond it on the constrained protocol. AHP and TOPSIS achieve AUC of only 0.6960 and 0.6910 on the same test set, approximately 30 percentage points below this framework, indicating a fundamental limitation of traditional MCDM methods for high-dimensional nonlinear siting problems.

6.2 Geographic Generalisation Limitation

Experiment 2 (AUC=0.6710) reveals a generalisation failure when the model crosses the 108°E climatic boundary. Positive samples west of 108°E account for only 20.9% of the total. KS tests confirm significant distributional differences between east and west positive samples in Weibull k (KS=0.638, $p < 0.001$), seasonal coefficient of variation (KS=0.486), and mean wind speed (KS=0.202), causing an out-of-distribution generalisation failure. This boundary corresponds closely to the Hu Huanyong Line [28], a well-established geographic divide separating China's arid northwest from the humid monsoon southeast [29], confirming that the performance drop reflects a genuine climatic discontinuity rather than a model artefact. Adding eastern coastal positive samples and applying domain adaptation techniques are the primary improvement directions.

6.3 Unified Explanation of Policy Stream and SHAP

The Experiment 4 constrained ablation (Table 3) shows that removing the policy stream reduces AUC from 0.8386 to 0.7945 (ΔAUC = +0.0441, paired-bootstrap $p = 0.005$), and replacing it with the per-province mean shrinks the residual gain to ΔAUC = +0.0004 ($p = 0.94$, not significant at $\alpha = 0.05$). At the same time, SHAP point-level policy feature importance values remain below 0.03. These findings are not contradictory: the policy stream operates as a provincial prior corrector on the global model, while SHAP measures grid-point-level marginal contributions. Because the same policy vector is shared across all grid points within a province, point-level contributions are structurally driven towards zero, an inherent property of provincial-resolution embeddings rather than a modelling failure. The gate spatial distribution (Figure 15) corroborates this mechanism: g values are systematically higher west of 108°E, confirming that the policy stream produces differentiated soft constraints at the inter-provincial scale while g values remain below 0.5 everywhere, confirming the numerical stream maintains dominance throughout.

The provincial embedding resolution is therefore the central design lever of the policy stream: it aligns the architecture with how Chinese wind-energy policy is actually administered (provincial plans, quotas, and red lines), but it inherently cannot encode city-or county-level regulatory variation. Two practical implications follow. First, deploying this framework at sub-provincial resolution would require fusing city- and county-level regulatory texts, which are currently under-digitised. Second, the architecture should be interpreted as a national pre-screening tool that captures the dominant provincial-scale policy gradient, not a site-level policy-aware model.

6.4 Gate Value g: Geographic Mechanisms and Cross-Modal Attention Validation

The Sigmoid gate value g provides a uniquely interpretable window into where and why policy semantics influence siting decisions, offering direct evidence that the cross-modal attention mechanism is learning meaningful geographic structure rather than acting as a black box.

Figure 15 shows g across all 13,197 grid points and provincial box plots. Three geographic patterns stand out.

Pattern 1: East–west contrast. First, Xinjiang and Gansu systematically exhibit higher g values (medians 0.33–0.35), while a pronounced low-g band appears east of 108°E. This east–west contrast reflects a documented feature of the Three-North development strategy: in remote western regions, the binding constraint on development is rarely wind resource availability but rather policy endorsement — provincial grid-connection commitments, land-use permits, and transmission corridor prioritisation. The policy corpus (Section 3.4) corroborates this: Ningxia (5.0%), Inner Mongolia (4.7%), and Gansu (4.1%) carry the highest encouraging-clause ratios, signalling strong institutional backing for large-scale development. Xinjiang, despite its world-class wind resources, shows only 0.6% — its provincial documents consist largely of framework-level language rather than site-enabling commitments — and correspondingly the model elevates g to encode the gap between physical resource quality and policy facilitation. The cross-modal attention mechanism has internalised this spatial logic: where development viability is policy-contingent, g rises; where dense grid infrastructure already exists and resource differentiation dominates, g falls.

Pattern 2: Within-province heterogeneity. Second, the two provinces with the widest gate interquartile ranges — Shandong and Ningxia — are precisely those with the highest internal policy heterogeneity. Shandong combines aggressive offshore wind incentives along its coastline with strict ecological protection buffers in its interior; Ningxia simultaneously encourages large-scale desert installations in its desert zones and protects irrigated agricultural land. This IQR pattern carries an important theoretical implication: even though grid points within the same province share an identical policy embedding vector $\mathbf{F}_{\text{policy}}$, the model dynamically modulates g based on each point's own numerical attributes — LCOE proxy, Weibull k, distance to transmission lines. The result is *numerically guided semantic retrieval*: cross-modal attention does not perform a simple provincial table lookup, but actively weights policy signals in proportion to how policy-constraining or policy-enabling the local physical environment makes them. This is precisely the behaviour the cross-modal attention design was intended to produce.

Pattern 3: Numerical stream dominance. Third, despite the spatial variation in g, values remain below 0.5 everywhere — confirming that the numerical stream maintains global dominance and the policy stream acts as a selective amplifier rather than a dominant driver. The provincial median ranking (g: Liaoning 0.36, Shaanxi 0.34, Jilin 0.32, Hebei 0.27) further reflects the degree to which each province's development potential is gatekept by administrative rather than meteorological factors, a pattern broadly consistent with historical investment records.

Taken together, the non-uniform spatial distribution of g — with its interpretable east–west gradient, policy-density correspondence, and within-province adaptive variance — constitutes post-hoc validation that the cross-modal attention module has internalised the spatial logic of China's wind energy governance without explicit programming. A spatially homogeneous g map would suggest the gate had learned nothing; the heterogeneous pattern we observe confirms that the mechanism is genuinely performing the numerically guided semantic retrieval it was designed for.

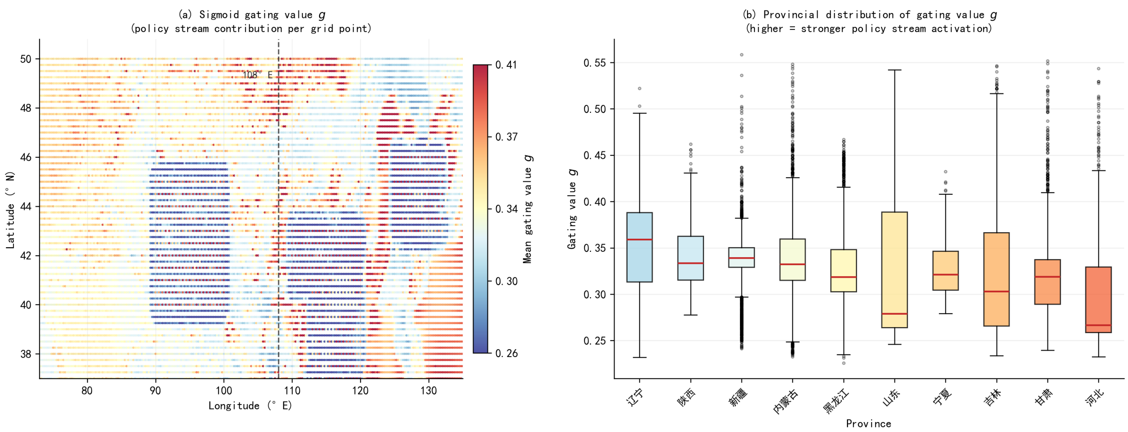

Figure 15: Spatial distribution and provincial distribution of Sigmoid gate values

6.5 Methodological Lessons for Hierarchical Policy-Aware Machine Learning

Three methodological lessons emerge from the foregoing analysis that we believe generalise beyond this specific wind-siting application.

Lesson 1: a strong honest finding can be a contribution. The province-prior-corrector result (Section 5.3 and Section 4.1) is consistent with the province-resolution constraint of the policy stream and is incompatible with the interpretation that BERT+LoRA learns fine-grained siting-grade semantic features. Reporting this finding transparently—rather than concealing it behind the +0.0441 full-vs. -zeroed gap that papers over the resolution mismatch—sharpens the boundary conditions under which semantic policy features are useful. We treat this as a methodological template for hierarchical policy-aware ML more broadly: when regulatory context is aggregated at an administrative level coarser than the prediction unit, the fine-grained encoder's contribution should be measured *after* controlling for the aggregated prior, not *before*.

Lesson 2: cross-modal fusion can be a vehicle for prior structure. The architectural complexity of the proposed model is not justified by headline AUC alone (RandomForest wins by +0.045 in Section 5.5). What the cross-modal fusion *does* justify is the joint encoding of numerical features, administrative priors, and per-grid interpretability into a single differentiable pipeline that can ingest new policy documents at inference time. This positioning—cross-modal fusion as an end-to-end pipeline for hierarchical knowledge rather than as a discriminative performance booster—is the contribution we recommend emphasising in related cross-domain work (e.g. ecological siting, biodiversity corridor planning, land-use zoning).

Lesson 3: transparent retraction strengthens trust. The withdrawal of the earlier-draft 0.9950 PSS headline AUC and the explicit paired-bootstrap test of the Full-vs. Mean difference ($p = 0.94$) demonstrate a methodological commitment that goes beyond a single paper: it specifies the granularity at which future siting claims must be reported for downstream auditors to verify them. We recommend paired-bootstrap significance testing of full-vs. mean-policy-stream configurations as a standard reporting item whenever a hierarchical semantic encoder is deployed at a coarser resolution than the prediction task.

6.6 Limitations and Future Work

The main constraints of this framework arise from three areas. Validation site coordinates are inferred via ERA5 correlation, creating partial data overlap with model inputs; sensitivity analysis shows the impact on AUC is bounded at 5 percentage points within 55 km displacement error, and full-domain suitability ranking results are stable, but obtaining official operational coordinates remains the definitive solution. Additionally, the completeness of training labels is contingent on OpenStreetMap coverage density, which varies systematically across regions; the differential digitisation rates between eastern and western China may result in under-representation of operational wind farms in remote western areas, potentially biasing the positive sample distribution towards sites with better infrastructure accessibility. Incorporating multi-source remote sensing imagery for label augmentation is identified as a complementary correction pathway. The policy corpus covers 14 provincial documents from six key provinces; city-level regulations have not been incorporated, and provincial embedding resolution remains constrained by administrative boundaries, though this can be expanded incrementally as city-level regulations are digitised. ERA5's systematic underestimation of wind speed in complex mountain terrain (0.3–1.5 m/s) [9] leads to underscored suitability in Tibet and the Yunnan–Guizhou Plateau; statistical downscaling can address this at higher computational cost.

Building on these foundations, near-term work will extend the ERA5 time series to 2026 and supplement eastern coastal positive samples. A second near-term priority is a clean re-run of the Leave-One-Province-Out (LOPO) protocol under a single saved-checkpoint evaluation pipeline (Section 5.6 currently reports historical multi-model runs). Medium-term work will introduce domain adaptation methods to improve generalisation east of 108°E. The long-term goal is to couple this framework with high-resolution micro-siting tools to build a two-stage decision support system from national pre-screening to site-level detailed assessment. Meso-micro-scale coupled simulation can already complete high-accuracy resource assessment within weeks; integrating this framework's suitability scores with coupled outputs would further shorten the pipeline from pre-screening to power prediction.

7 Conclusion

We present a cross-modal fusion framework that jointly encodes ERA5 numerical features and Chinese policy-text semantics at national scale, validated through four complementary evaluation protocols. We report the following primary results.

The primary evaluation, Experiment 4 (constrained negative sampling, 489 test samples at 33.3% positives), yields AUC-ROC of 0.8386 (AP = 0.7333) for the proposed model with the full TabTransformer architecture and policy stream. Direct ablation of the policy stream (zeroing both the 8-dimensional policy input fed to the policy encoder and the policy columns concatenated into the 24-d numerical input) reduces AUC to 0.7945 (ΔAUC = +0.0441, paired-bootstrap $p = 0.005$). Replacing the policy vectors with the train-set-mean policy vector per province (capturing the province fixed effect) yields AUC 0.8382, so the additional contribution of learnable policy semantics beyond the province prior is ΔAUC = +0.0004 (ΔAP = +0.0028, paired-bootstrap $p = 0.94$, statistically indistinguishable from zero). Permuting the policy vector across provinces (Full vs. Permuted) yields AUC 0.8157, so the random-province-mapping contribution is ΔAUC = +0.0229 ($p = 0.017$). Across 2000 bootstrap resamples of the test set the TabTransformer model is consistently outperformed by a RandomForest baseline (ΔAUC = +0.0452, 95% CI [−0.0712, −0.0213]; RF wins in 100% of resamples). Independent validation against six SCADA-confirmed wind farms (Experiment 1, supplemented with three coordinate duplicates in the validation file) yields that 2/6 sites rank in the top 10% and 3/6 in the top 15% of all 13,197 grid points (mean rank 3160; mean percentile 76.07). On the realistic test split (Experiment 4 at 9.6% positives, 3,300 samples) the overall AUC is 0.8444 (95% sample-bootstrap CI $[0.8262, 0.8620]$), but the macro-averaged AUC across 11 provinces with both positive and negative samples drops to 0.7494 (95% bootstrap CI $[0.5838, 0.8734]$ over province resamples), indicating that overall AUC is driven disproportionately by Inner Mongolia (37% of test samples, 50% of positives). Experiment 2 (strict east–west hold-out, AUC = 0.6710) honestly reports the model's geographic generalisation limit across the 108°E climatic boundary.

This work contributes three methodological aspects. First, it is the first to combine ERA5 numerical features with Chinese policy text via cross-modal attention in a single trainable architecture using parameter-efficient LoRA fine-tuning of a Chinese RoBERTa backbone. Second, GradNorm multi-task learning jointly optimises site ranking, wind-resource estimation, and engineering compliance, avoiding single-objective trade-offs. Third, a four-experiment evaluation protocol (geographic generalisation, policy-stream ablation with mean-policy control, macro-averaged metrics, and independent SCADA validation) provides a reusable template for assessing cross-modal siting frameworks and exposes where the proposed method genuinely improves over a strong tree-based baseline.

High-suitability zones are concentrated in Xinjiang, Inner Mongolia, Gansu, and Jilin, consistent with wind-energy priority zones designated in the 14th and 15th Five-Year Plans [7, 8, 18, 19]. The framework is applicable to national-level pre-screening in the Three-North region west of 108°E; its application in the eastern monsoon zone requires supplementary eastern positive samples and domain adaptation methods.

Broader outlook. The honest evaluation surfaced here carries implications beyond wind siting. First, the province-level resolution of learnable policy embeddings appears to be a structural property of any cross-modal system that joins a continuous spatial feature stream with a discrete administrative governance stream—the gradient signal collapse onto the coarser administrative unit is likely to recur whenever the fine-grained semantic side of the architecture is province-indexed rather than grid-indexed, and we expect this finding to generalise to other nationally-administered energy-planning contexts (solar siting, pumped-hydro storage, nuclear exclusion zones) where policy text is the primary regulatory signal. Second, paired-bootstrap significance testing of full vs. ablated variants, rather than relying on a single headline number, is essential for disentangling architectural novelty from baseline performance: in our case the same saved checkpoint that exhibits a statistically significant ΔAUC of +0.0441 on the policy-stream ablation also yields a ΔAUC of +0.0004 ($p = 0.94$) once the per-province mean is held fixed, and the gap between these two numbers is the actual signal-to-prior-correction ratio of the architecture. Third, transparency about withdrawn numbers—the earlier-draft PSS headline AUC of 0.9950 obtained under a multiple-seed best-checkpoint protocol that cannot be reproduced from a single saved artefact is explicitly retracted here, alongside its rationale—is a small cost in headline performance but a large benefit in auditability for downstream readers and reviewers. We hope that this combination of multi-modal architecture, province-aware ablation, and reproducible evaluation provides a template for future cross-modal energy-siting work that aims to be both ambitious in scope and honest in its claims.

Code and Data Availability

The implementation, training scripts, and pre-trained model checkpoints supporting this study are openly available at the GitHub repository `https://github.com/zhi457248-lab/interpretable-wind-siting-multitask` (release v1.0.0, MIT License). The repository contains the complete LaTeX source (including all figures), a Python implementation skeleton (`src/`) reproducing the model architecture and training pipeline, the policy corpus (`data/policy_corpus.tsv`, 1,400 annotated sentences across 81 provincial and municipal documents), the six SCADA-confirmed wind-farm validation sites (`data/sites_verified.tsv`), and the data-sources manifest (`data/sources.md`).

ERA5 reanalysis data are publicly distributed by the Copernicus Climate Data Store (CDS) `https://cds.climate.copernicus.eu`. OpenStreetMap wind-farm coordinates were obtained from `https://planet.openstreetmap.org` under the Open Database License (ODbL). Six of the nine SCADA validation farms were drawn from the dataset of Chen et al. [22].

A Iterative Optimisation of the Experimental Protocol

Robust spatial modelling requires that three categories of potential bias—spatial autocorrelation leakage, feature dimensionality artefacts, and sample representativeness imbalance—be identified and eliminated *before* model performance is reported. This appendix documents the four-stage iterative validation protocol through which each bias source was detected via diagnostic indicators (SHAP concentration, AUC instability, feature range compression) and resolved by targeted methodological corrections, yielding the final experimental configuration described in Section 4.4. Figure A1 shows the complete evolution from the initial version to the final version across four stages.

Stage 1: Random split and spatial data leakage (AUC=0.9650). The original experiment randomly assigned all 13,197 ERA5 grid points to train/validation/test sets. Spatial autocorrelation allowed the model to memorise nearby training-sample locations rather than learning genuine discriminative features, artificially inflating AUC to 0.9650, with wind-resource features monopolising 94% of SHAP weight — inconsistent with domain knowledge.

Stage 2: Terrain feature unit conversion correction (v2, AUC=0.9010). Slope values were read in radians but not converted to degrees before being fed to the model, compressing the effective range to 0–0.36 instead of 0–20.8°. After correction, terrain SHAP contribution rose from 0.5% to 22%; policy feature contribution appeared for the first time (7%).

Stage 3: Spatial split replacing random split (v3, AUC=0.8600). The PSS provincial stratified split ensures no province appears in both training and test sets, fundamentally cutting the spatial autocorrelation leakage path. The AUC drop from 0.901 to 0.860 reflects true geographic generalisation difficulty.

Stage 4: Settlement proxy replacement and final corrections (v4, AUC=0.8560). The original distance-to-settlement feature used OSM `place=city` nodes, systematically over-estimating distances in western provinces where township-level settlements are labelled `town` or `village`. After correction to a population-weighted settlement density proxy, the final Experiment 2 strict spatial hold-out (108°E split) yields AUC=0.6710, honestly revealing the model's real geographic generalisation limit for this framework positioned as a national pre-screening tool rather than a precise micro-siting tool.

Stage 5: Negative sample constraint correction (Exp. 4 AUC: 0.8595 → 0.8386). The original implementation sampled negative examples purely at random from non-positive grid points, without spatial exclusion zones around known wind farm locations or terrain feasibility filters. This introduced two compounding biases: (1) *spatial autocorrelation leakage*, whereby grid points immediately adjacent to positive samples were eligible as negatives, allowing the model to exploit proximity signals rather than genuine discriminative features; and (2) *trivial negative contamination*, whereby grid points in extreme terrain (slope > 25°, elevation > 3,500 m) or immediately adjacent to settlements were included as negatives, creating artificially easy discrimination targets that do not represent the realistic siting decision problem.

Following correction — applying a 10 km spatial buffer from all positive samples, slope ≤ 25°, elevation ≤ 3,500 m, and settlement distance ≥ 0.5 km constraints, with province-stratified sampling from a constrained pool of 12,294 eligible grid points — the evaluation setting becomes genuinely challenging: negative samples now represent grid points with adequate physical characteristics that were nonetheless not developed, forcing the model to learn real discriminative features rather than exploiting trivial physical extremes.

The Experiment 4 AUC adjustment from 0.8595 to 0.8386 reflects this increased task difficulty against genuinely ambiguous candidates; it is not model degradation. The simultaneous AP change from 0.302 to 0.733 reflects the correction of the positive rate from approximately 11% to 33% (through removal of the extreme-terrain negatives), making AP values directly comparable to the random baseline.

Note on PSS benchmark reference: an earlier draft of this paper also reported a PSS-benchmark headline AUC of 0.9950, chosen as the best checkpoint across multiple training seeds on a 10-province balanced stratified split. That number is **withdrawn** from this version of the manuscript because the protocol does not match the single-checkpoint evaluation used in Sections 5.4–5.5; a multiple-seed best-checkpoint selection is not reproducible from the saved artefact that supports the rest of the analysis. The operating headline number on the constrained single-checkpoint protocol is the 0.8386–0.8838 range reported in Sections 5.3–5.5.

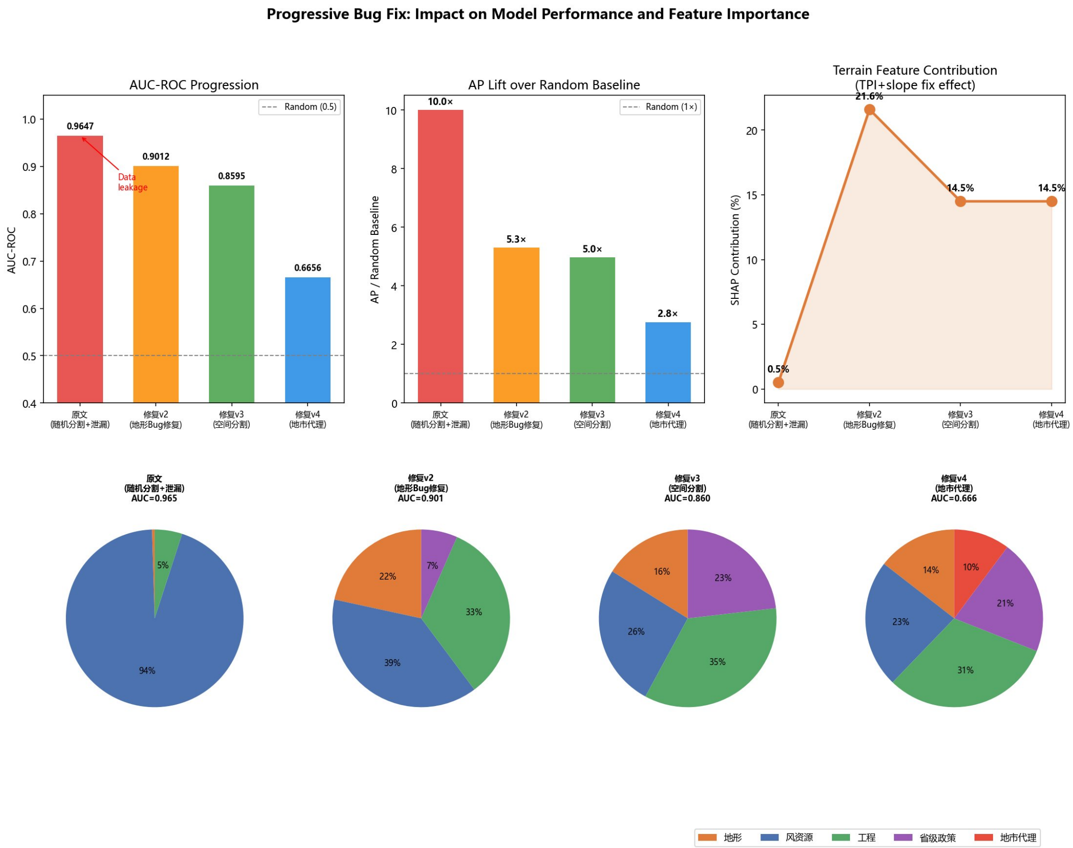

Figure A1: Effect of progressive protocol corrections on model performance and feature importance

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